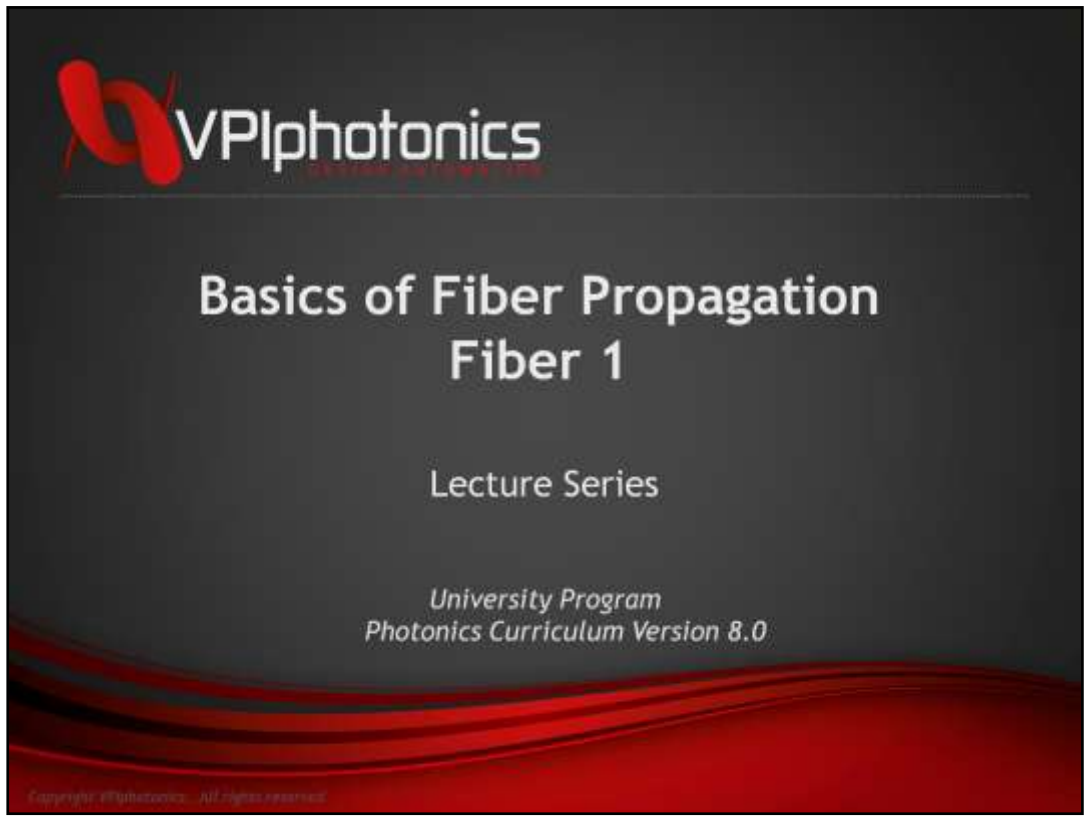
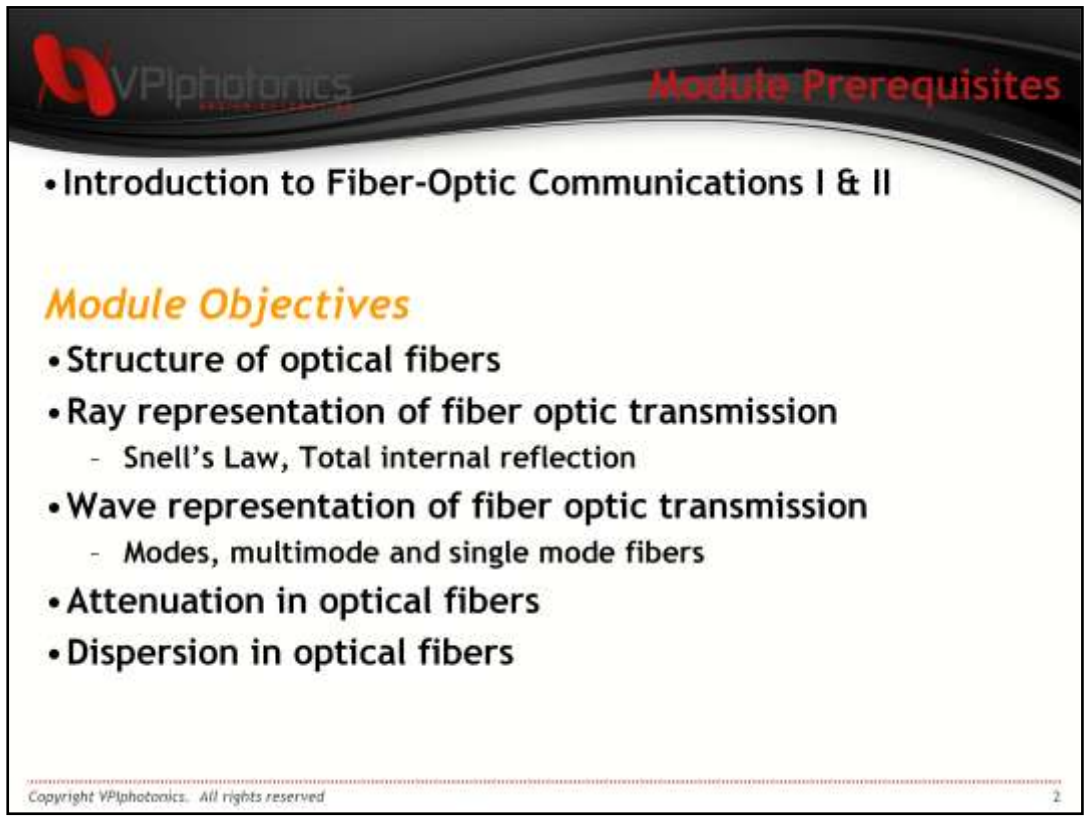
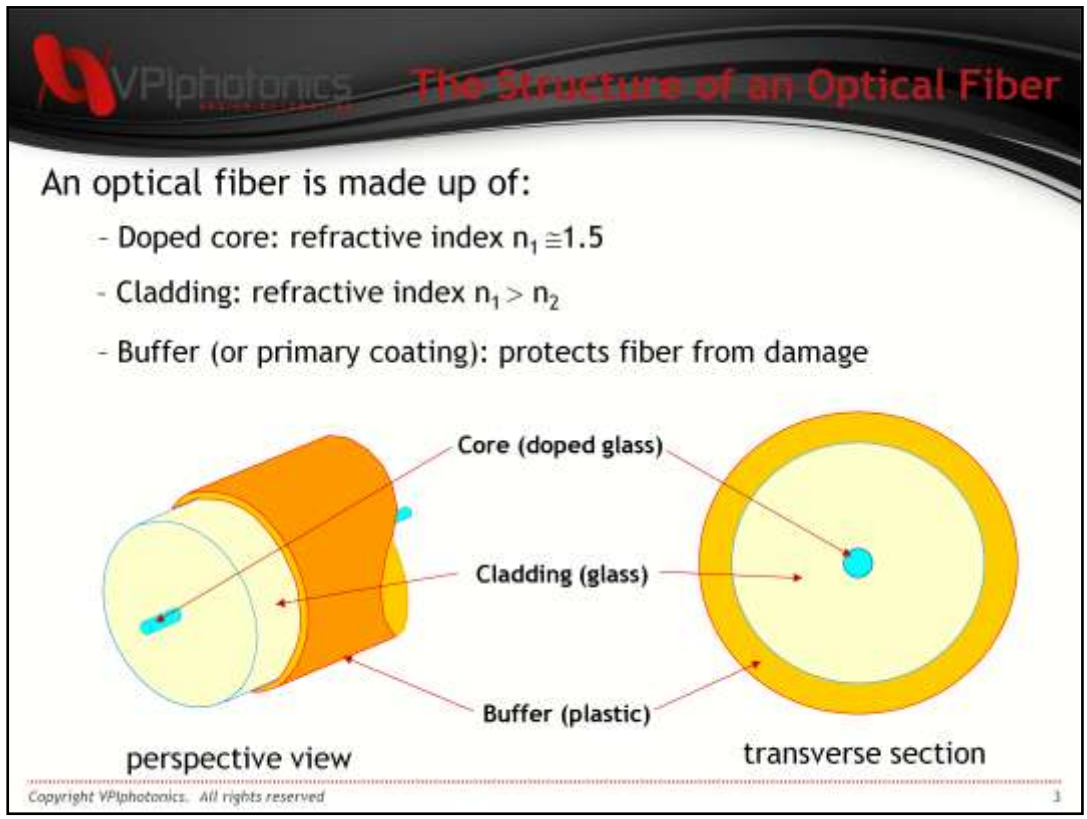

VPIphotonics
Basics of Fiber Propagation
Fiber 1
Lecture Series
University Program
Photonics Curriculum Version 8.0



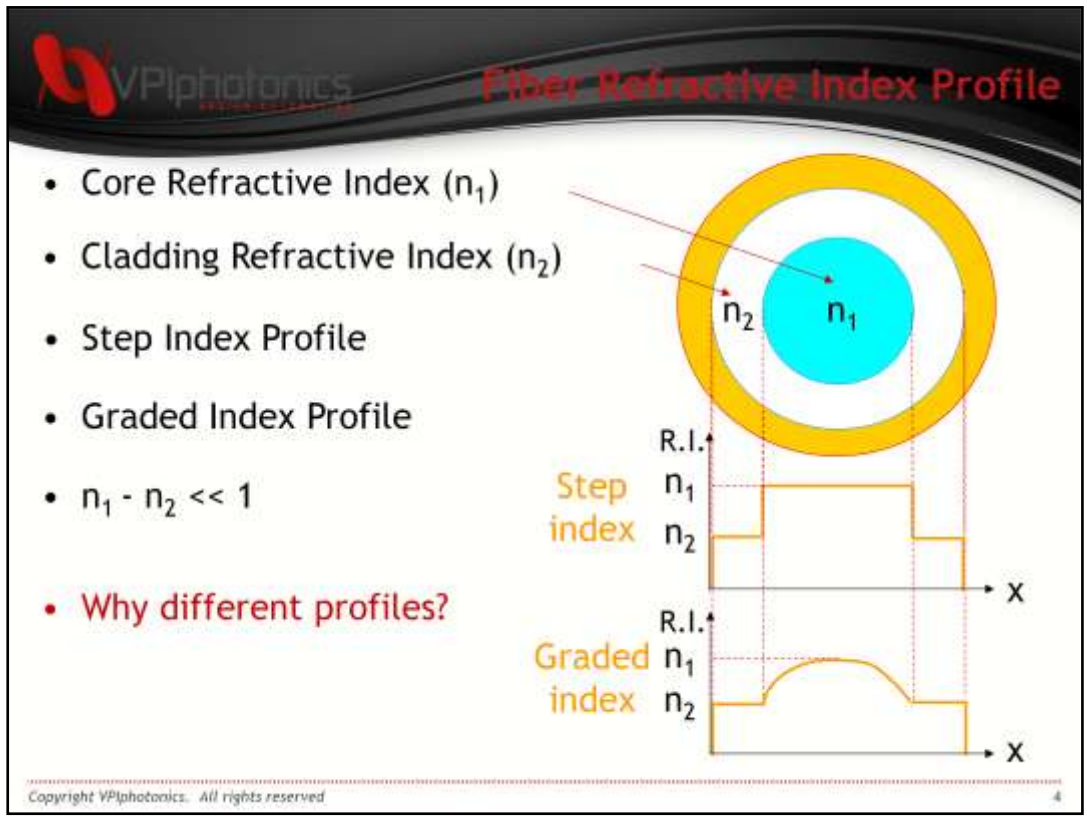
Module prerequisites and objectives – edit to suit the specific requirements of your course.



An optical fibre consists of:

- A core made of doped silica (glass)
- A cladding that surrounds the core, made of (undoped) glass
- A plastic jacket or buffer that protects the fiber from damage and also helps to strengthen it.

The refractive indices of the core and cladding are different, with the core having a slightly higher refractive index than the cladding. This, as will be explained later in the lecture, is the important characteristic of optical fibers that enables them to transmit light.



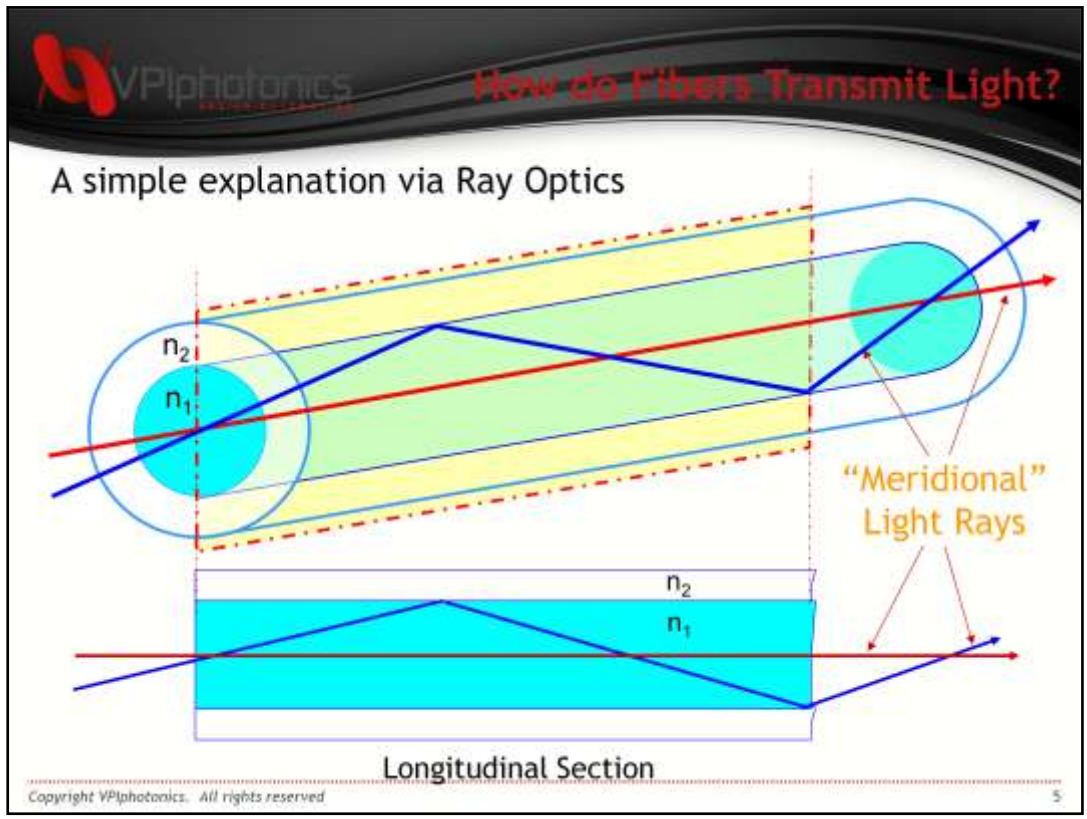
The doped core has a refractive index n_1 this is very slightly higher than the cladding refractive index n_2.

The difference in refractive index, $(n_1 - n_2) << 1$, but this small difference is extremely important, and forms the whole basis of how the transmission of light through optical fibers.

In general, there are two types of Refractive index profile (the variation of the refractive index of an optical fiber as a function of its radius):

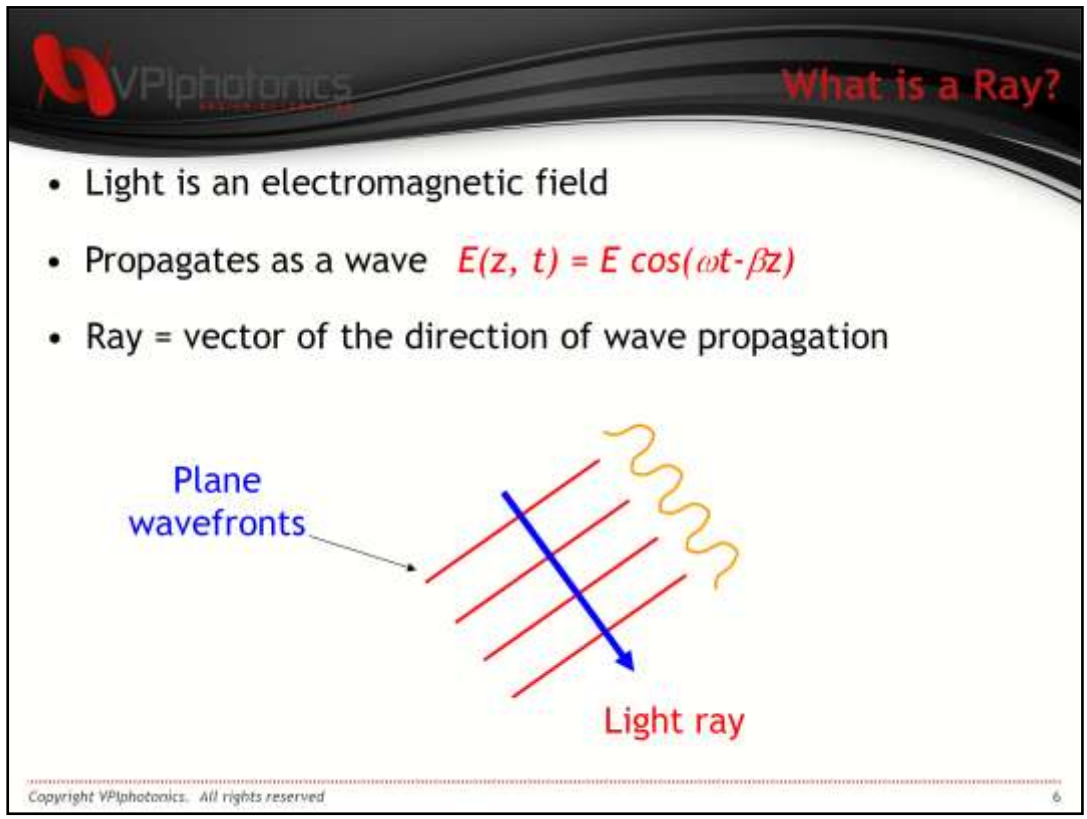
- Step index, where the refractive index changes abruptly
- Graded index, where the refractive index changes gradually from the center of the core to the interface boundary between the core and the cladding

There is a reason why different profiles are used – this will be explained in another slide later.



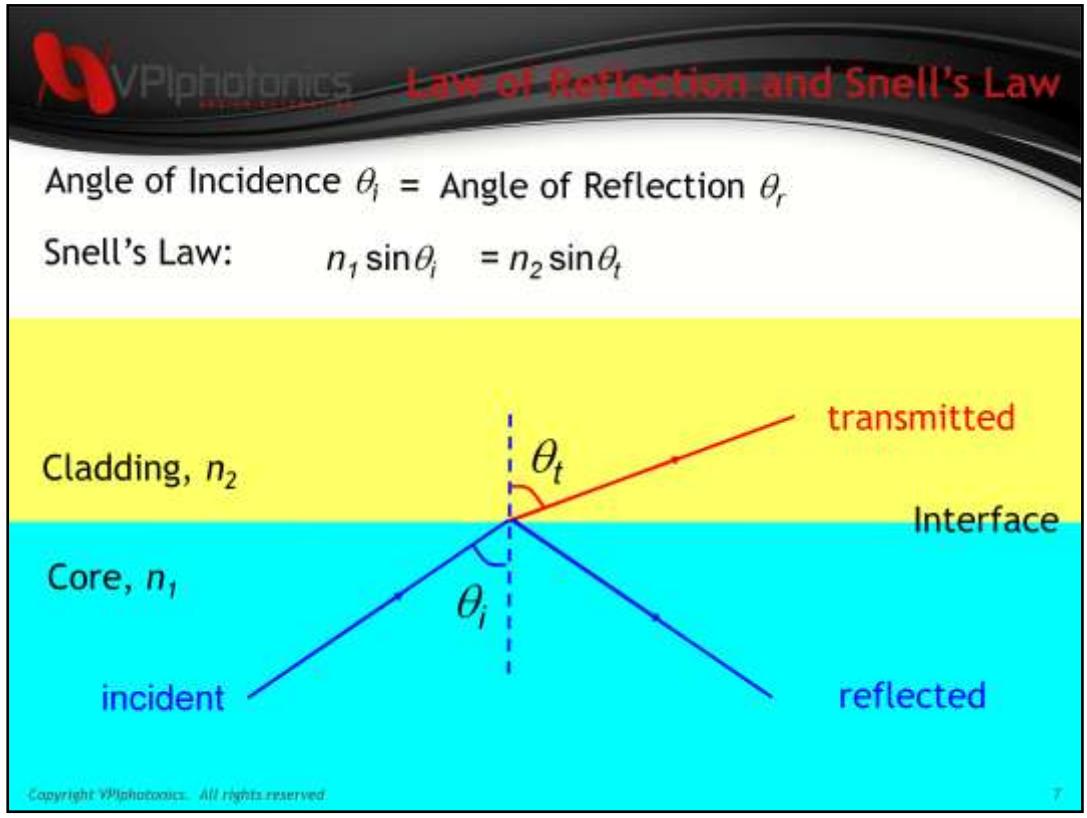
A simple picture of how optical fibers transmit light can be given using Ray Optics.
A light ray can propagate through a fiber through internal reflection, as shown by the orange ray above.
It is assumed that these rays are "meridional", i.e. they pass through the center or axis of the fiber core.
Other rays that do not pass through the fiber axis are possible (these are "skew" rays), but we don't need to consider them here in order to introduce the conceptual picture of light propagation in optical fibers.
The mathematical analysis can be facilitated by taking a longitudinal section through the center of the fiber.
The longitudinal section looks like the cross-section of a two dimensional planar waveguide.
Because of the difference in the refractive indices between the cladding and the core, one can apply Snell's Law to analyse the path of the light rays.

Note that this simple ray picture is really only valid when the dimensions of the core are much larger than the wavelength of the light, so the effects of diffraction can be neglected. For practical optical fibers, the ray picture is highly "conceptual" (i.e. incorrect!)

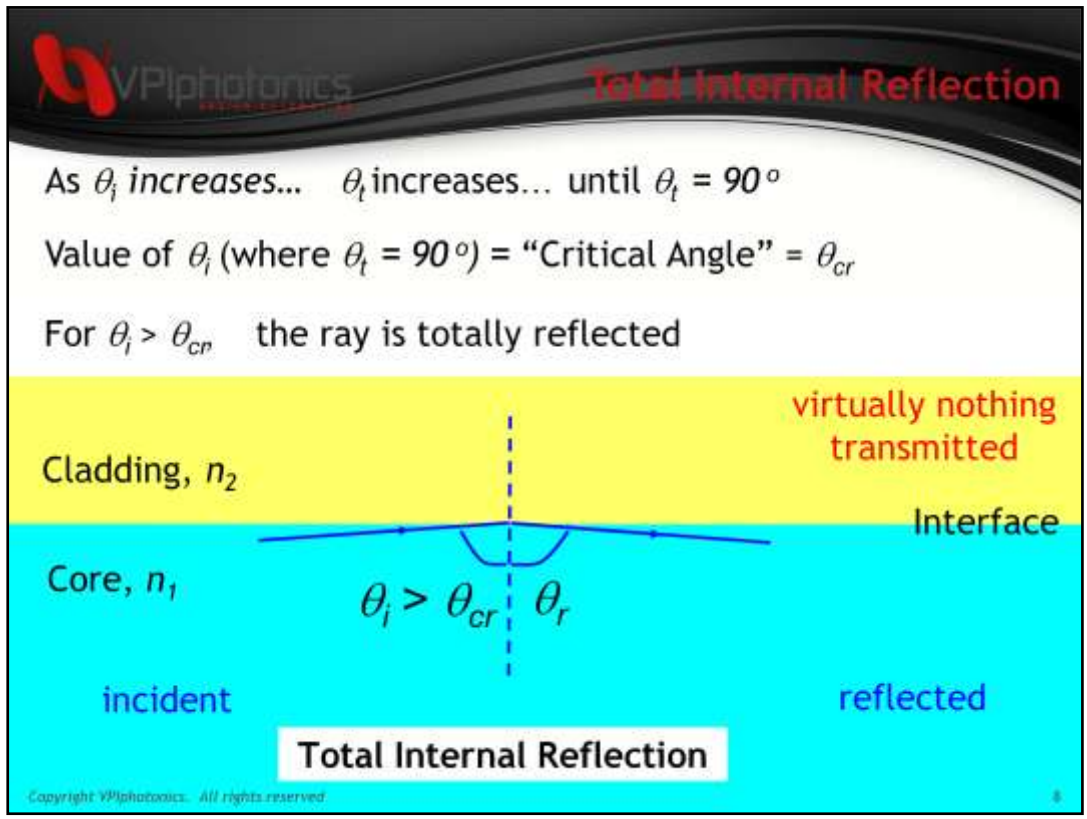


Light, like other electromagnetic fields, propagates as a wave, so what does a “ray” represent?

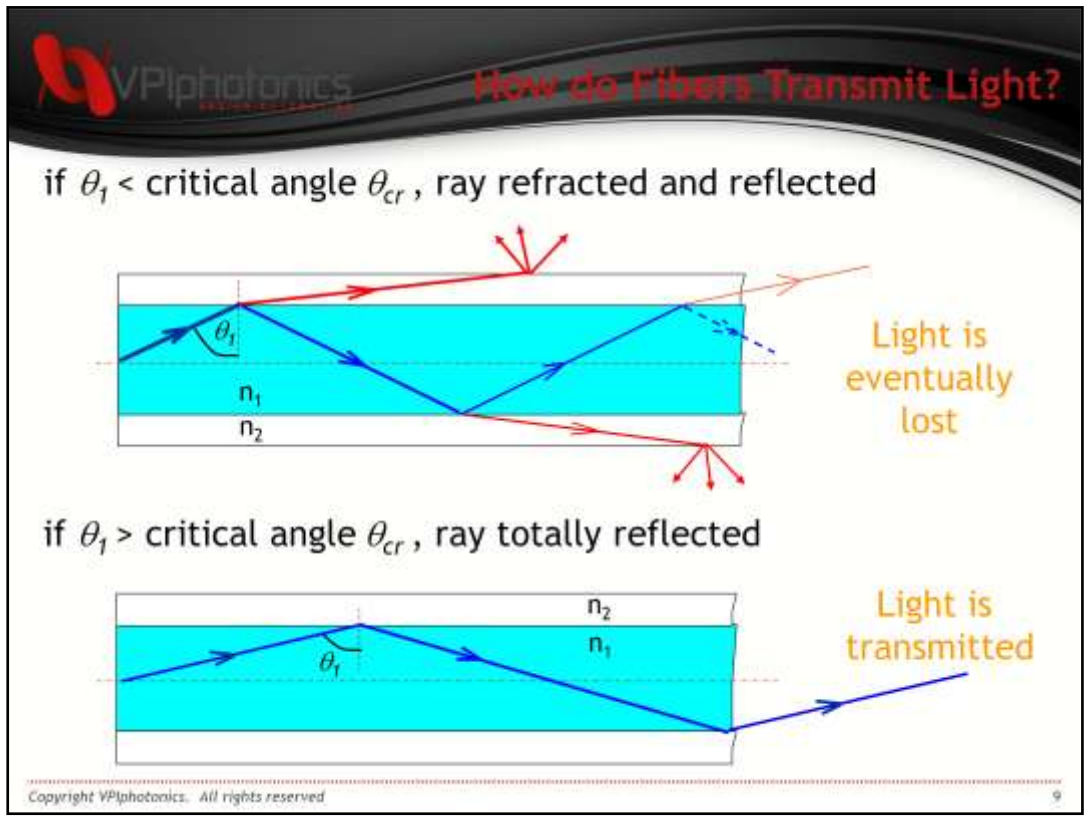
Well, a ray can be understood to be a vector representing the direction of propagation of a plane wave, i.e. one in which all the wavefronts are parallel to each other, and perpendicular to the direction of propagation.



When a ray of light strikes an interface between two materials of different refractive index, some of the light is transmitted and some of it is reflected.
The angle of incidence Theta_i equals the angle of reflection Theta_r.
Snell's Law relates the angle of incidence Theta_i to the angle of transmittance Theta_t.

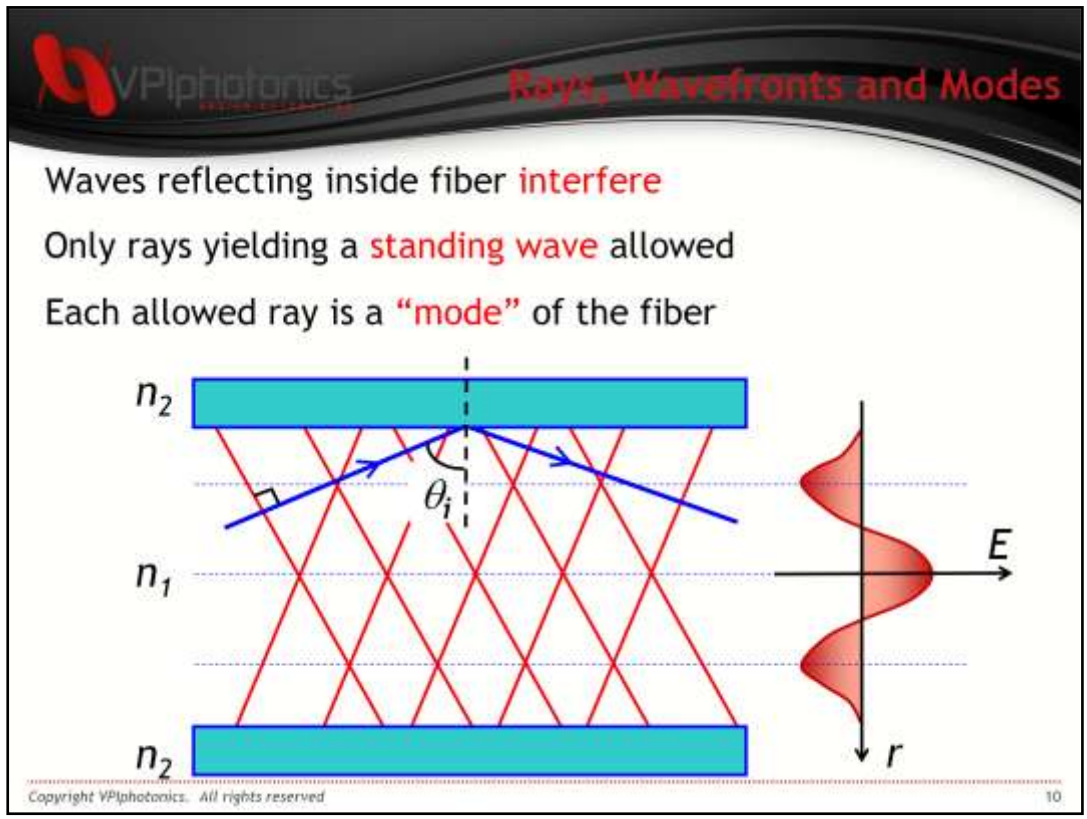


As the angle $\Theta\eta\varepsilon\tau\alpha_i$ increases, the angle of transmittance $Theta_t$ increases, until at the angle $\Theta\eta\varepsilon\tau\alpha_t = 90^o$, in which case virtually nothing is transmitted. The value of $\Theta\eta\varepsilon\tau\alpha_i$ for which $\Theta\eta\varepsilon\tau\alpha_t = 90^o$ is known as the critical angle – the ray is totally reflected. This is known as Total Internal Reflection, and is the basis by which light is transmitted through an optical fiber.



For values of Theta_i < the critical angle, the ray is loses some energy each time it is reflected off the boundary between the core and cladding – eventually, the ray is lost as all its energy is lost.

For values of Theta_i > the critical angle, the ray is totally reflected and propagates down the fiber with minimal loss (except those incurred as a consequence of propagation in the core material).



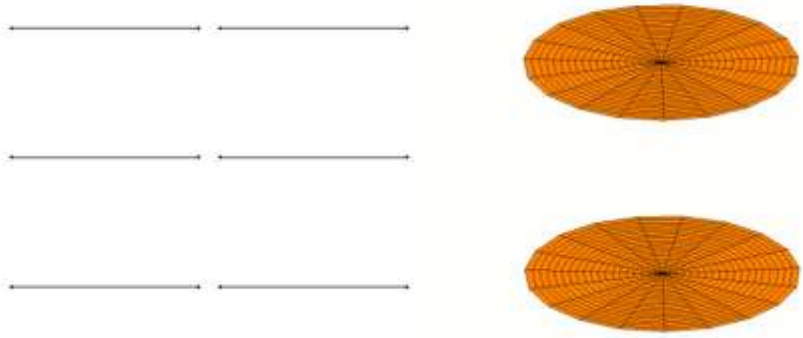
Consider the waves represented by a ray reflecting inside a fiber core.

These waves must interfere inside the core:
- where two peaks (or troughs) meet, we get a large peak or a trough
- where a peak and a trough meet, we get cancellation

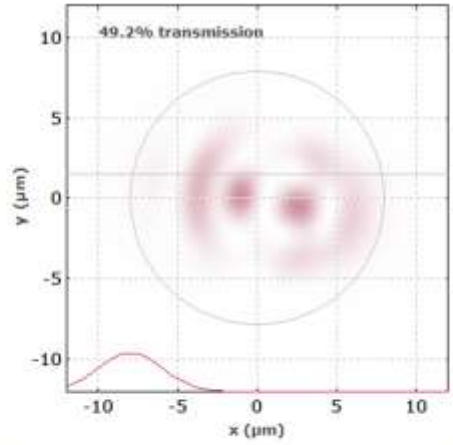
Only rays, whose interfering waves result in a standing wave pattern across the fiber cross-section, are allowed to propagate through the fiber. Show in the diagram is the standing wave in the Electric Field distribution in the transverse direction (across the fiber cross-section).

Each such allowed ray corresponding to a single or unique "mode" of the fiber (or waveguide in general).

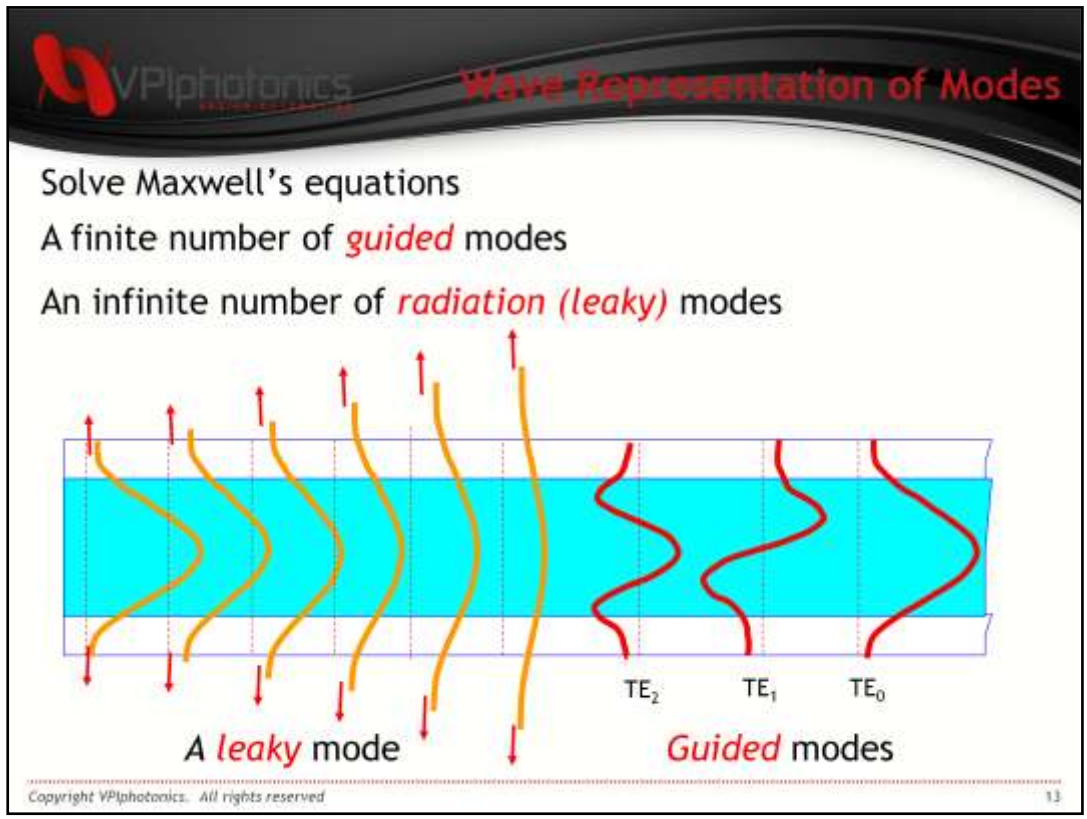
- In physics, a **standing wave** - also known as a **stationary wave** - is a wave in which its peaks (or any other point on the wave) do not move spatially.
- The locations at which the amplitude is minimum are called nodes, and the locations where the amplitude is maximum are called antinodes.



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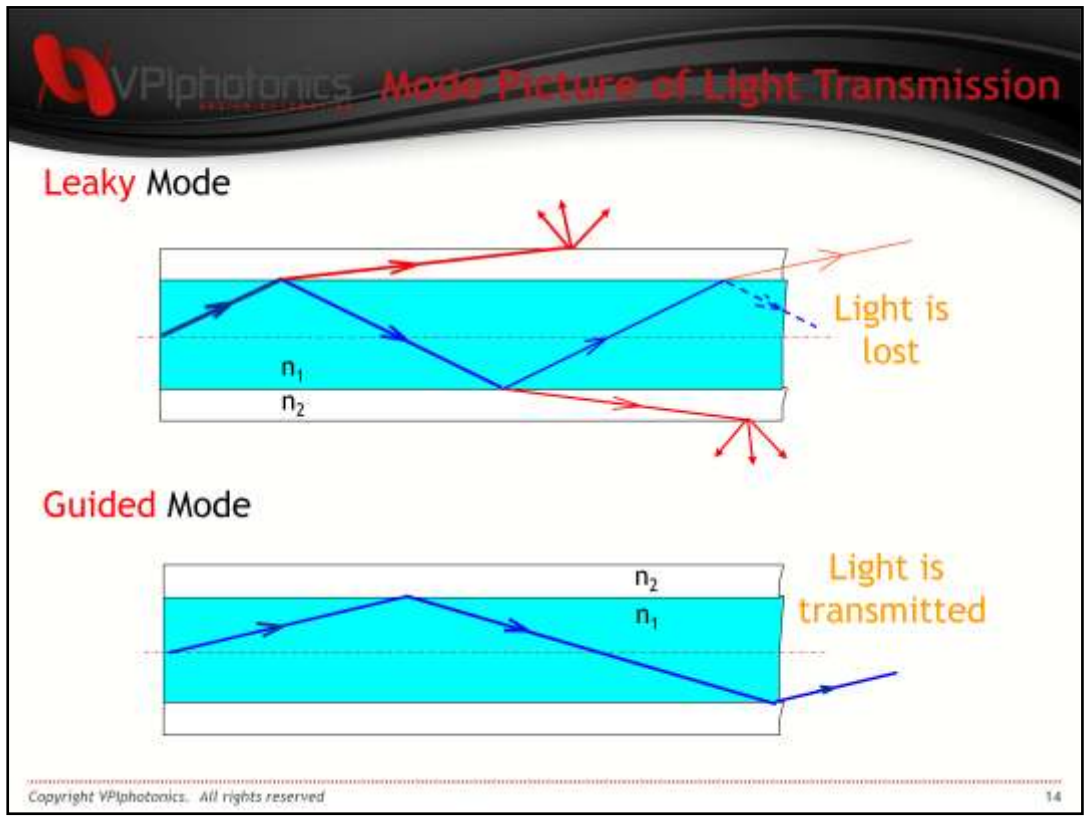


A full understanding of the way in which light is guided in an optical fiber requires the solution of Maxwell's equations to determine the "modes" of the waveguide.

Solving Maxwell's equations shows that the optical fiber supports a finite number of *guided* modes and an infinite number of *radiation* or *leaky* modes.

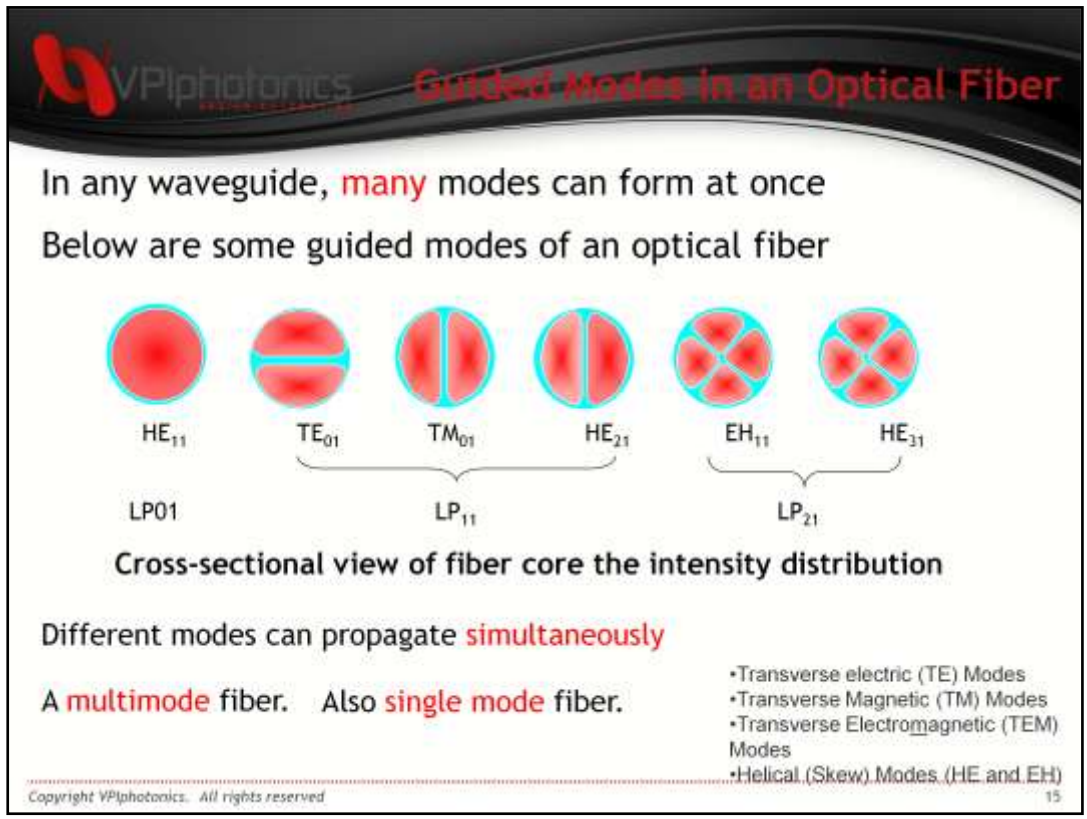
Guide modes propagate through the fiber will little loss (except that caused by the material properties of the fiber and bending).
Leaky modes lose energy from inside the core to the cladding and beyond. As such, they do not propagate through the fiber as they are quickly radiated away from inside the fiber due to the leakage of energy. The leakage of energy occurs as sinusoidal waves outwards from the cladding/air boundary – this is not drawn in the diagram above – the mode outside of the cladding should be sinusoidal to show the radiation of energy away from the cladding – and the amplitude of this sinusoid decreases with distance travelled by the mode in the fibre (the standing wave of the mode inside the core also decreases in amplitude as energy is drained from it by the sinusoidal waveforms outside the cladding).

As an example, Maxwell's equation is solved for a comparatively simple waveguide, that of a symmetrical slab waveguide (whose longitudinal cross-section is identical to that of an optical fiber). The mathematics is fairly simple, but it serves as a good preparatory exercise for the more difficult solution of Maxwell's equations for optical fiber with a circular transverse cross-section.



This picture is only a ray representation of what happens in the fibre. The leakage of energy in a leaky mode occurs as sinusoidal waves outwards from the cladding/air boundary – this is not drawn in the diagram above. – the mode outside of the cladding is sinusoidal to show the radiation of energy away from the cladding (in practice, this radiation hits the opaque protective jacket covering the cladding and is absorbed/lost). The amplitude of this radiation decreases with distance travelled by the mode in the fibre. Likewise, the standing wave of inside the core also decreases in amplitude as energy is drained from it by the radiative wave outside the cladding).

All this are revealed by the mathematical derivation of the light propagation in a step index optical fiber, which is not really that difficult for 3rd or 4th year students to do. Perhaps this can be set as an exercise.

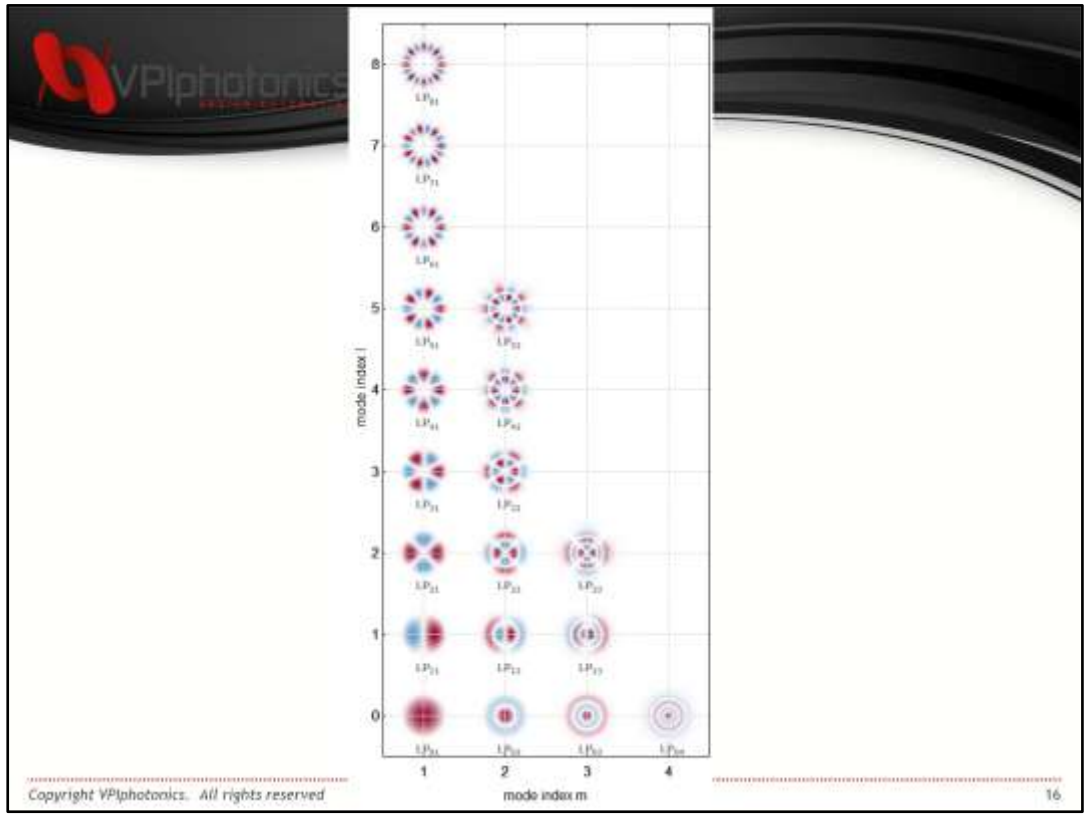


Multiple modes form in any waveguide situation

Shown in this slide are some of the lowest order modes of an Optical fiber with circular cross-section and a step refractive index profile:

- Transverse electric (TE) Modes
- Transverse Magnetic (TM) Modes
- Transverse Electromagnetic (TEM) Modes
- Helical (Skew) Modes (HE and EH)

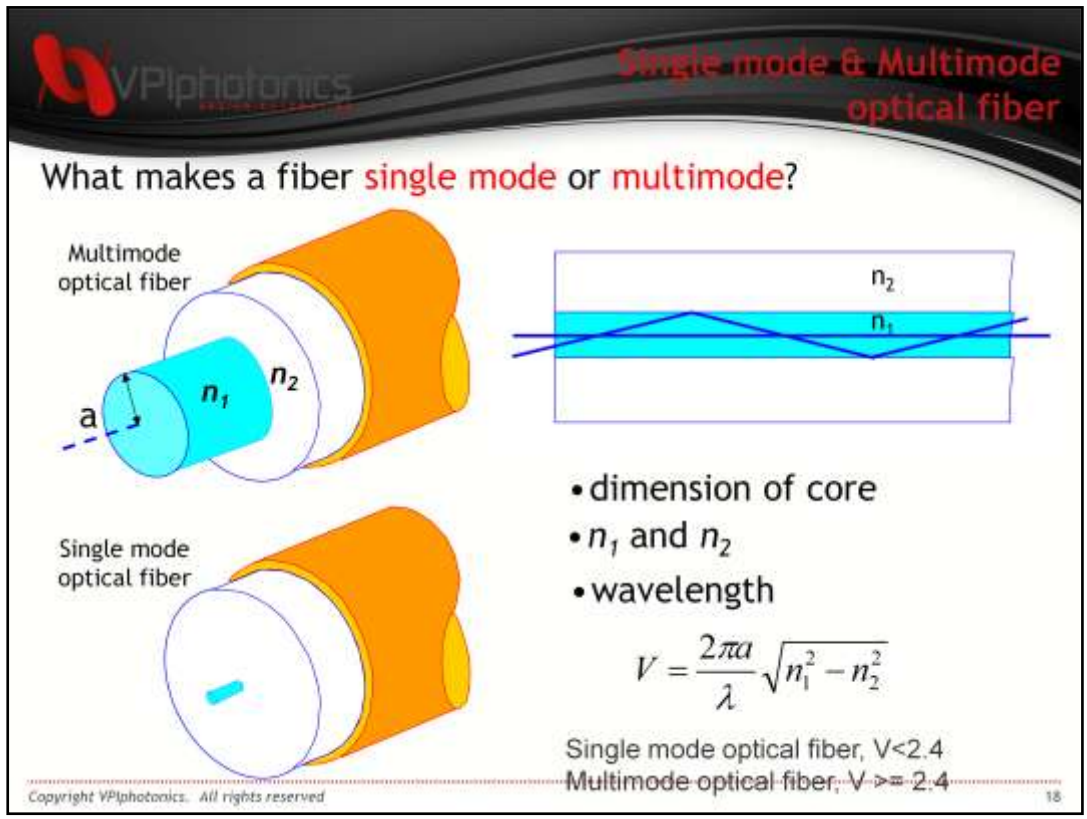
Modes with the same transverse intensity distribution (but are made up of different electromagnetic field combinations) are usually grouped into *Linearly Polarized* (LP) Modes, as shown in the slide.



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The number of modes that can propagate simultaneously in an optical fiber depends mainly on the wavelength of the light, the dimensions (radius) of the fiber core and the refractive indices of the fiber core and cladding.

In solving Maxwell's equation for an optical fiber, a dimensionless parameter V is defined and used to characterize the guiding properties of an optical fiber.

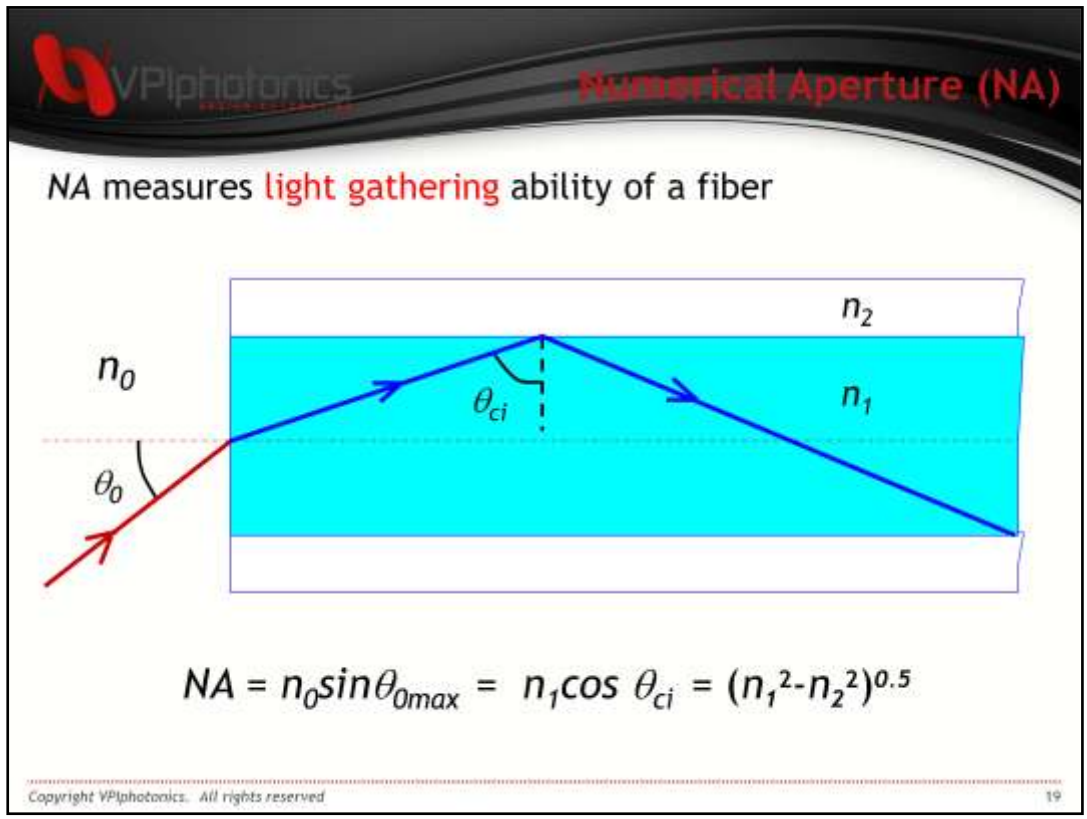
V = (2*a*πι/λαμβδα)∗[(ν1⊥2−ν2⊥2)⊥(1/2)],
ωηερε α = χορε ραδιυσ, ν1= χορε ρεφραχτιϖε ινδεξ, ν2 = χλαδδινγ ρεφραχτιϖε ινδεξ, λαμβδα = ωαϖελενγτη οφ λιγητ

It can be shown that for:

- Single mode optical fiber, $V<2.4$
- Multimode optical fiber, $V >= 2.4$

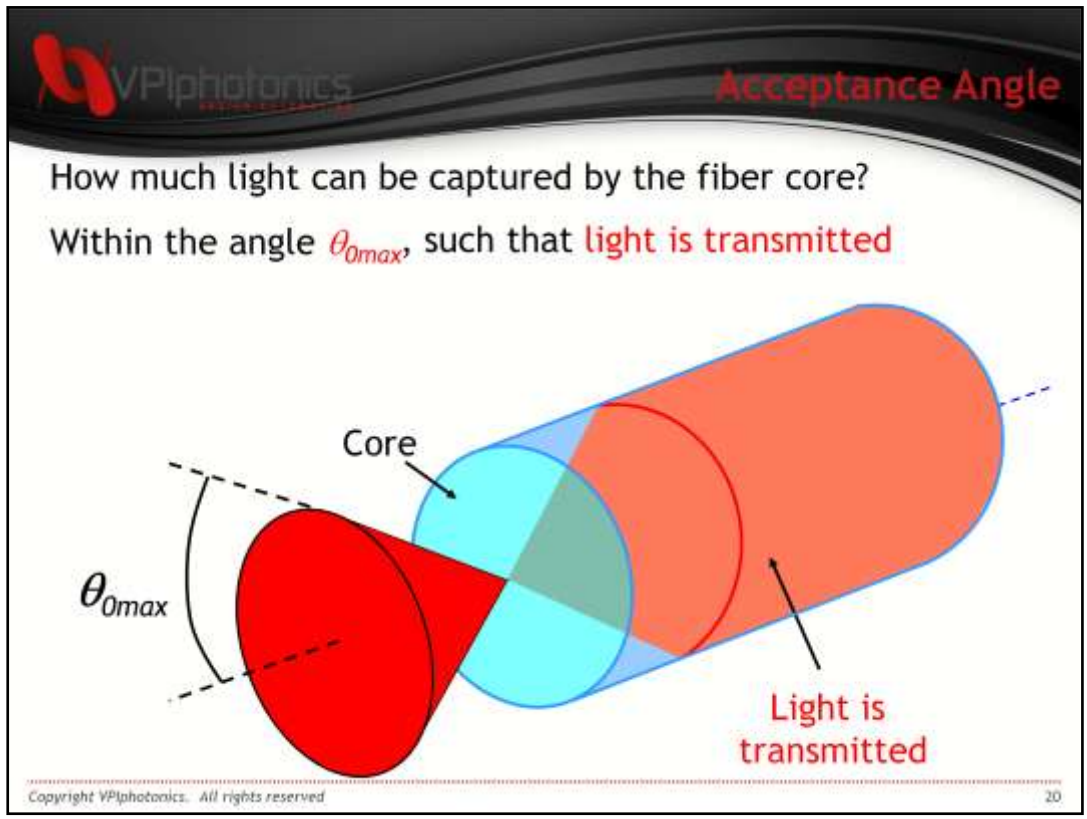
The number of guided modes M in a fiber:

-Step index fiber, $M = V^2/2$
-Graded index fiber, $M = V^2/4$



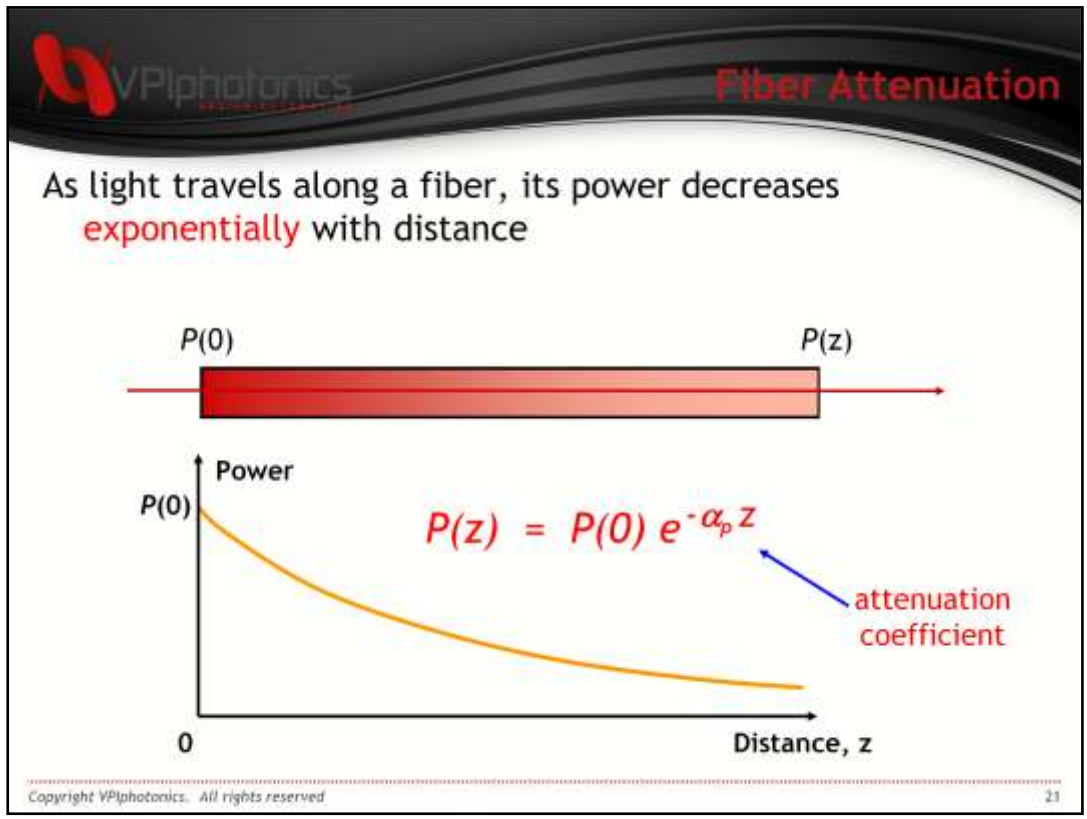
Typical values of NA:
Single mode fiber: *NA* = 0.1 ~ 0.2
Multimode fiber: *NA* = 0.2 ~ 0.5

Challenge the students to derive the NA equation, as a take home or tutorial exercise.

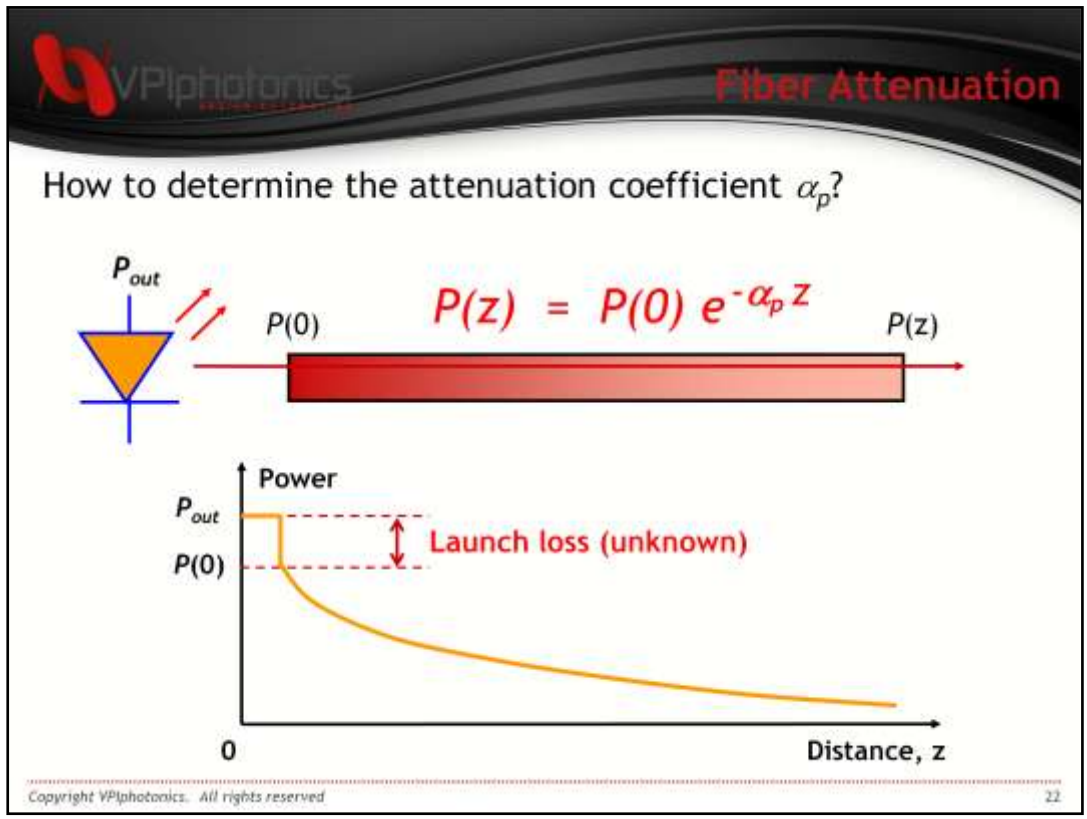


Typical values of Acceptance Angle:
Single mode fiber: θ_{omax} 3.8° ~7.7°
Multimode fiber: θ_{omax} 7.7°~ 20°

The obvious ramification of the acceptance angle is that light from an external source is difficult to couple into an optical fiber.

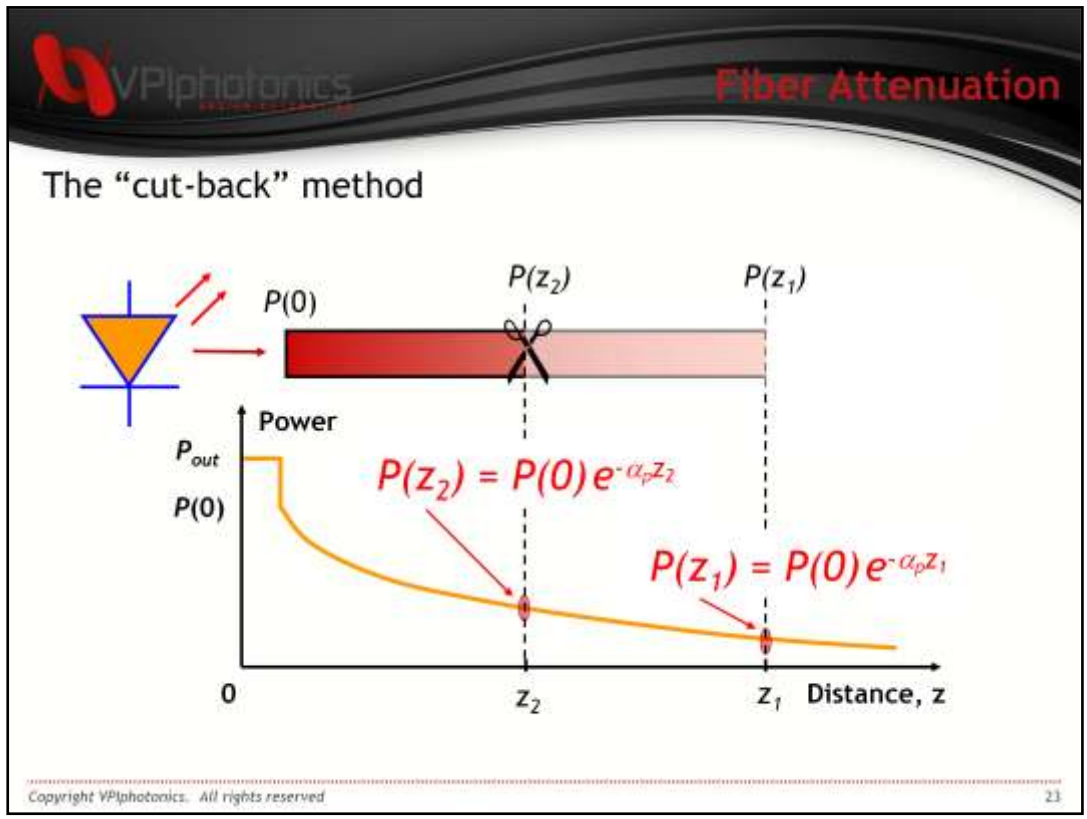


Fiber attenuation causes the power level of a light wave to decrease exponentially with distance propagated. The rate of decrease is determined by the attenuation coefficient, $alpha_p$, of the fiber.



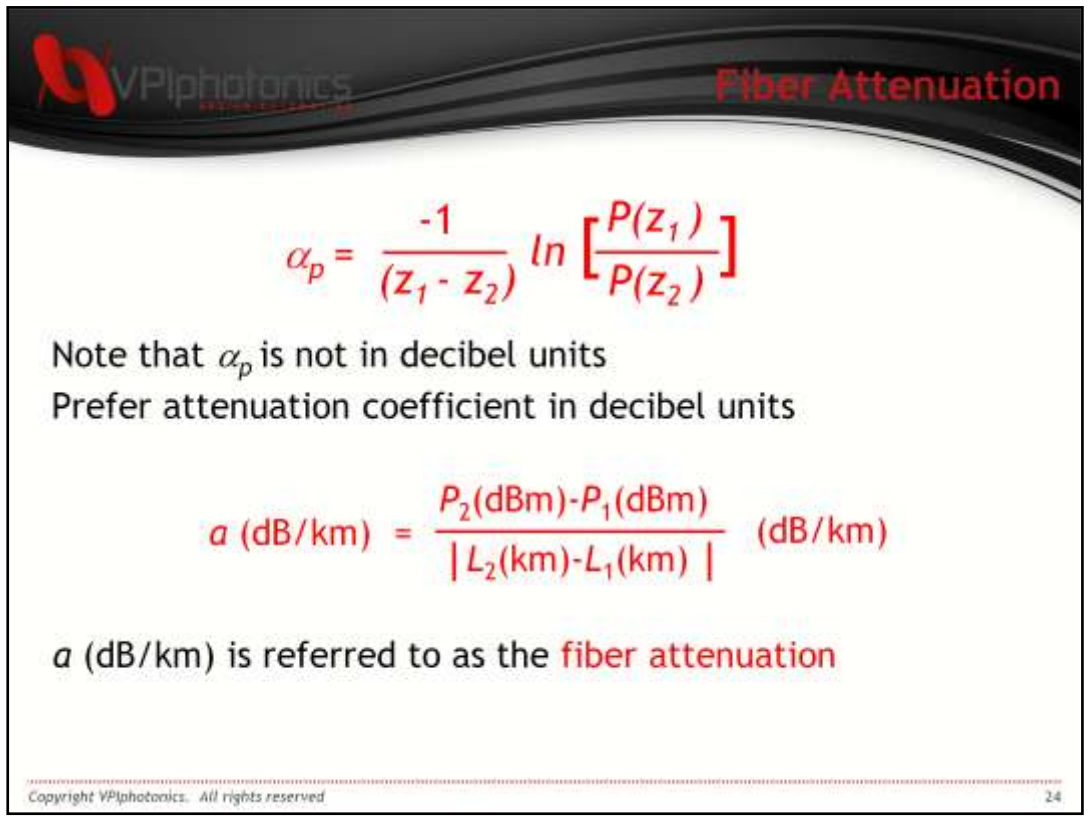
Challenge the students to figure out how to determine the attenuation coefficient in practice.

Note: P(0) is difficult to measure because of the launch (coupling) loss (between light source and the fiber core) is not known exactly.



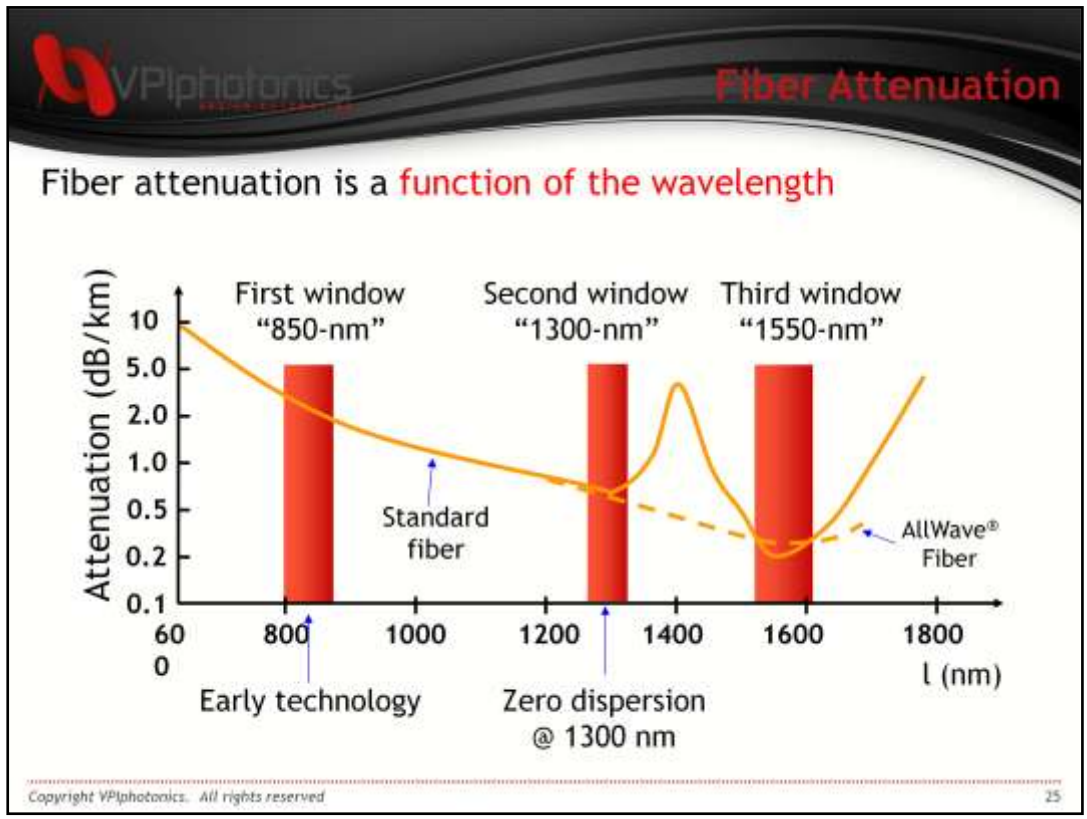
Here is how it is done in practice, using the "cut-back method".

The power is measured at some (known) distance z_1 from the input. Then, the fiber is cut, reducing its length to z_2 (distance measured from the input). Then a couple of simple mathematical steps (which the students should be able to perform), and the answer can be found (or see the next slide).



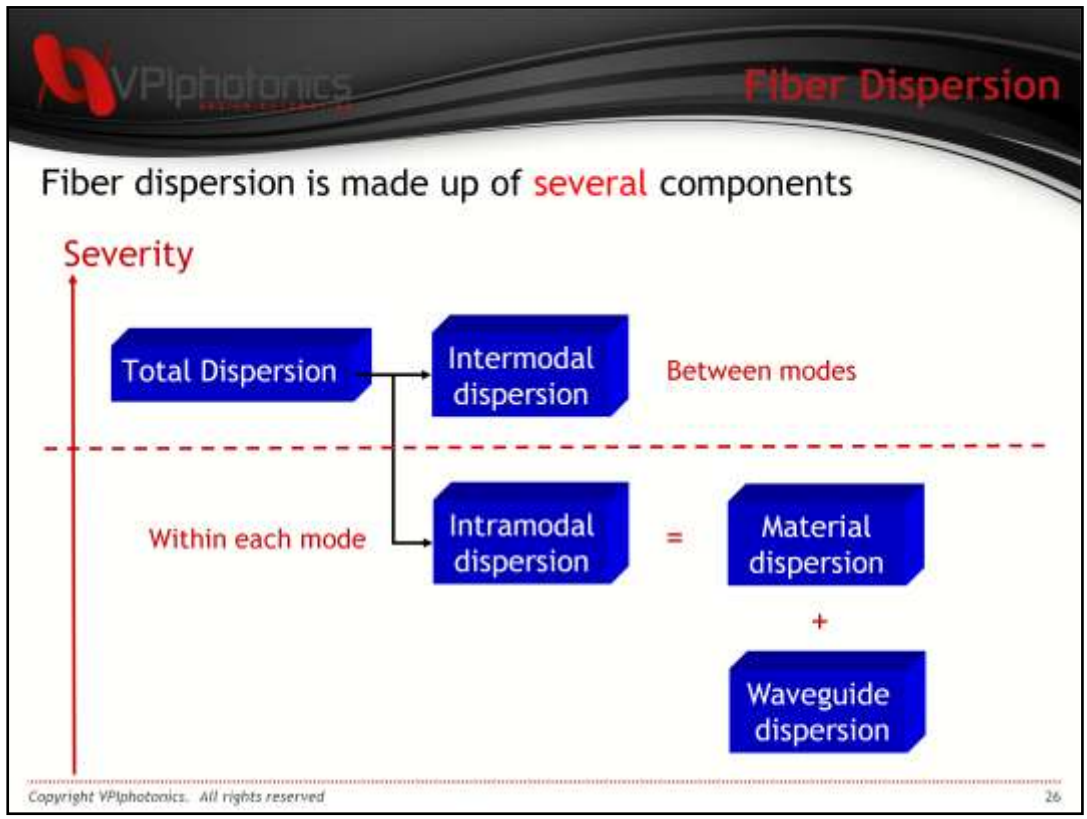
The simple mathematical steps leads to the expression above (simply plug in the measured values and the known values).

Note that the attenuation coefficient thus calculated is not in decibels (which is the preferred unit, and what is found in datasheets).

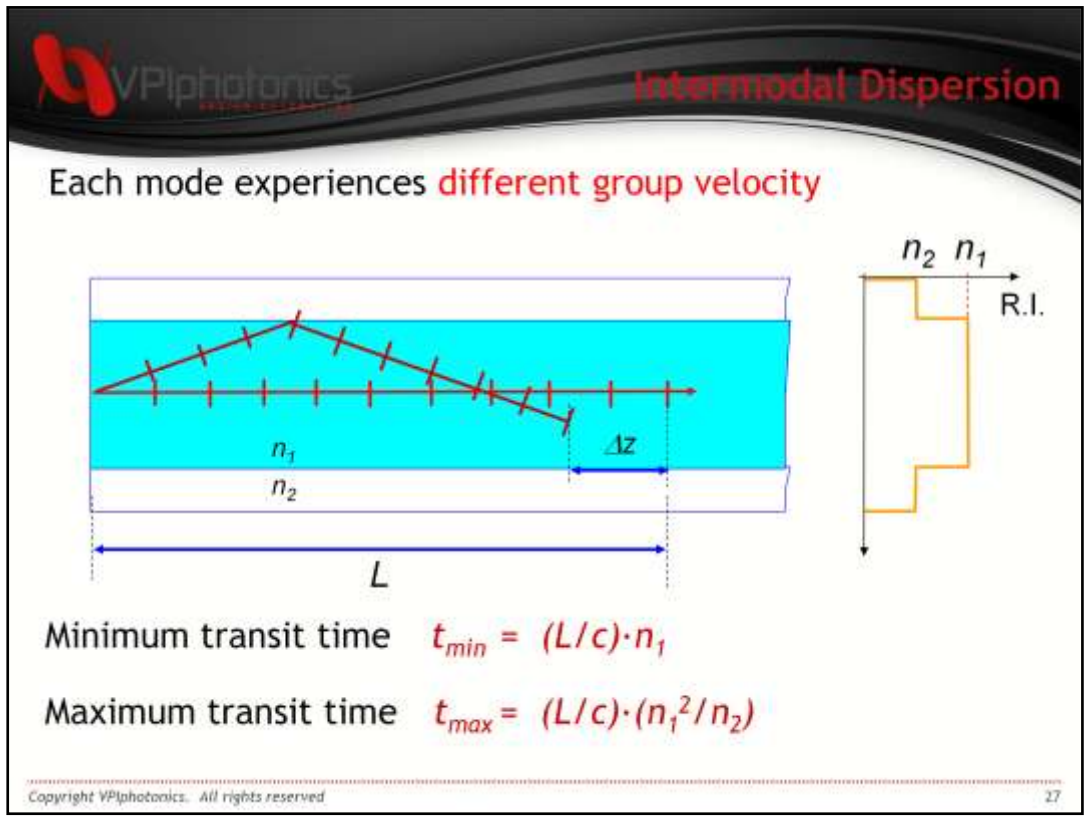


The peak in the attenuation vs wavelength curve are cause by OH- (hydroxyl ions) impurities in the glass. It is also commonly referred to as the “water peak”.

New fiber design and manufacturing processes have reduced or eliminated the OH- attenuation peak from the curve.



Fiber dispersion has been introduced in the introductory modules. It is made up of several components, as depicted in the slide. Total dispersion = intermodal + intramodal dispersion. Intramodal dispersion in turn is made up of material and waveguide dispersion.



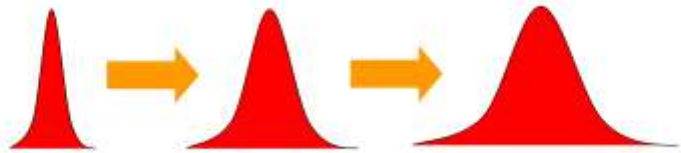
Intermodal dispersion: a result of each mode having a different value of the group velocity at a single frequency.

Firstly let's consider a step-index fiber:
Considering the requirement for total internal reflection, the minimum and maximum transit times are given by:
Tmin = (L/c).n1
Tmax= (L/c).((n1^2)/n2)

Intermodal Dispersion

The delay difference or pulse spread in time:

$$\delta t_{mod} = t_{max} - t_{min} = (L/c)n_1(n_1/n_2 - 1) \cong (L/c)(NA^2/2n_1)$$

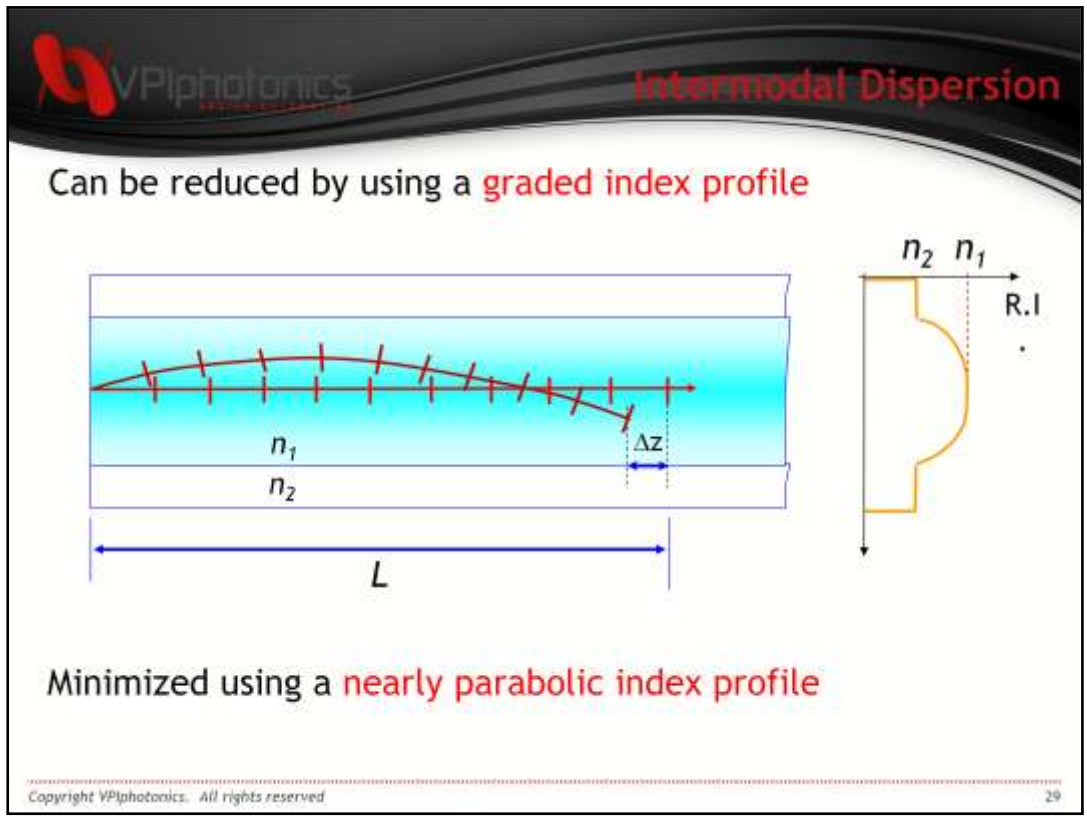


The rms pulse broadening per unit length due to intermodal dispersion (for a step index fiber):

$$\sigma_{mod} \cong \frac{(NA)^2}{4\sqrt{2}n_1 c} \quad \text{ns/km}$$

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The different in transit times of the different modes result in the pulse eventually spreading apart (also known as “broadening”).



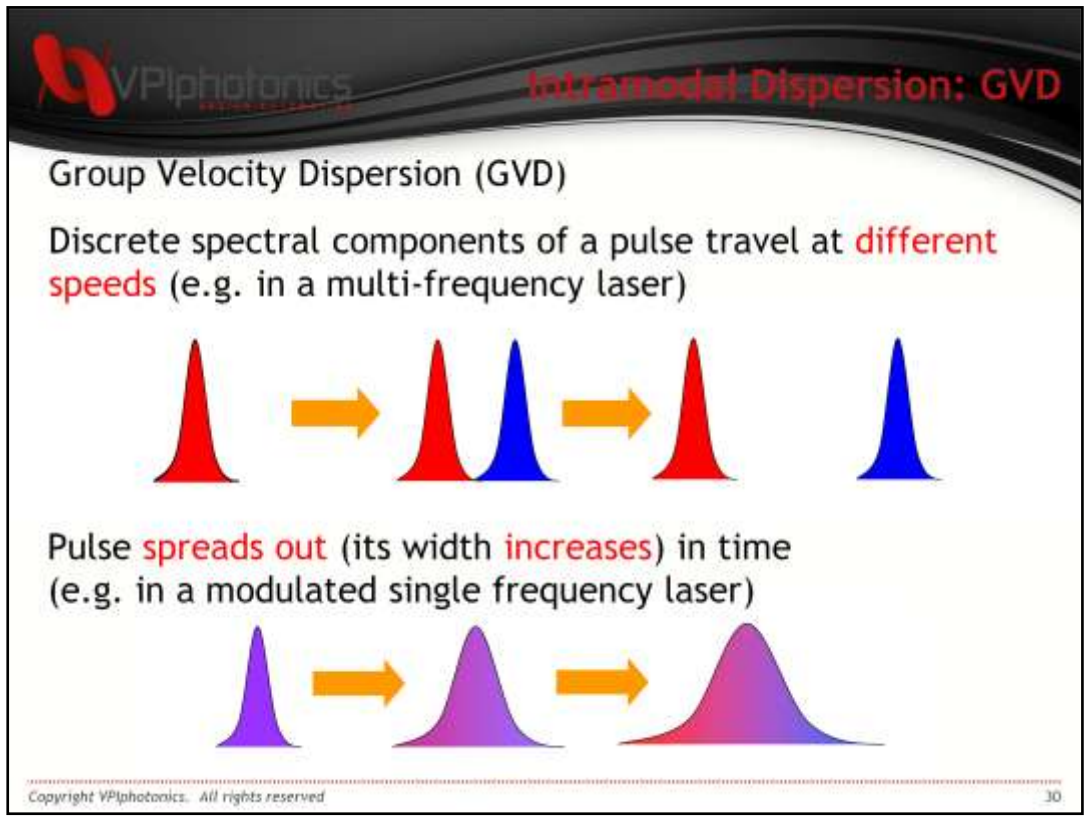
The pulse broadening due to intermodal dispersion may be reduced by grading the index of the core

Compared to the axial ray, the curved ray travels through material with a lower (average) refractive index, thus its moves faster (on average). Thus, the time difference is reduced.

Is there a profile that is “the best”, in terms of reducing or eliminating intermodal dispersion? Yes, there is an optimum graded index profile where the intermodal dispersion is reduced to a minimum.

It has been found that this optimum profile is very nearly parabolic.

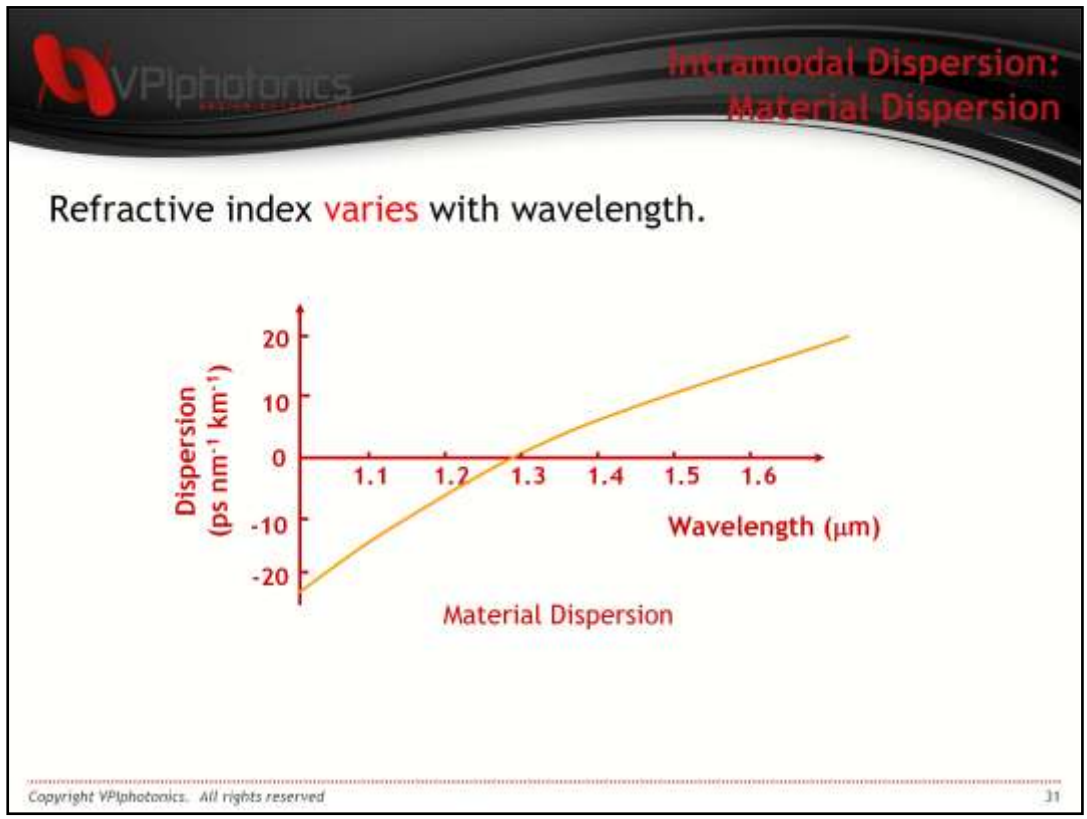
Note also that intermodal dispersion can be avoided by reducing the dimensions of the core, to the point that only one guided mode can propagate in the core. Since only one mode exists, intermodal dispersion is also eliminated.



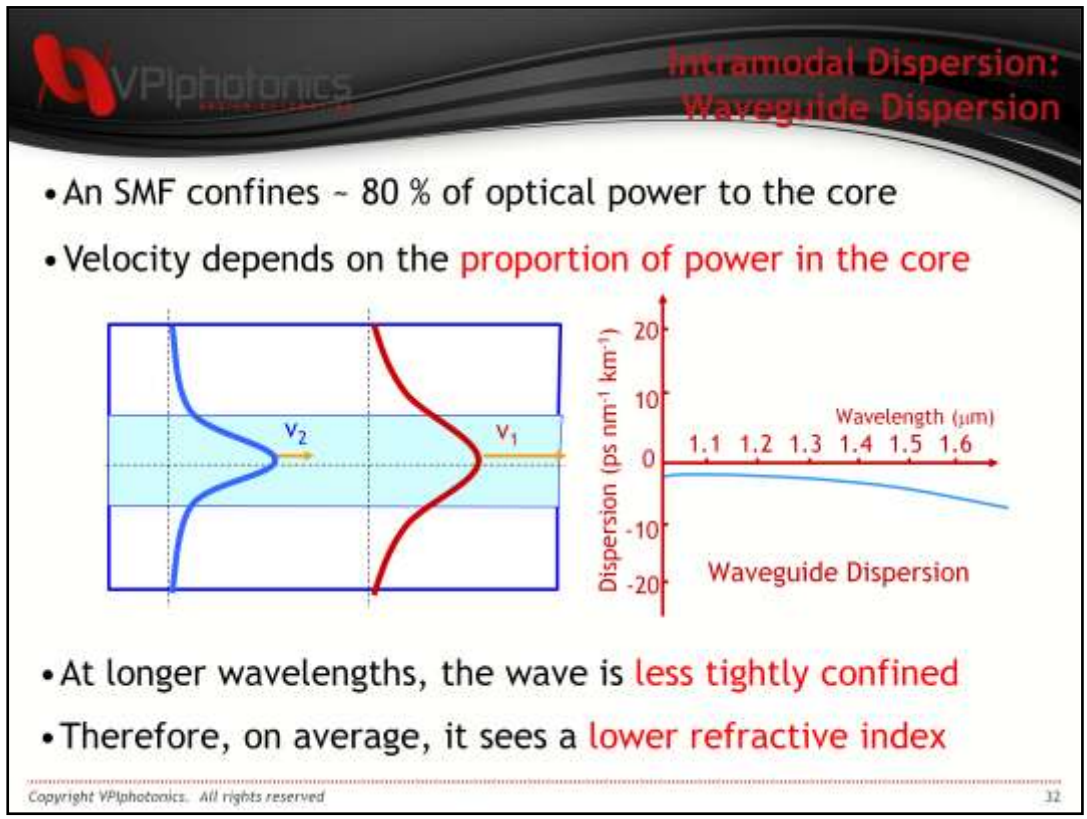
Considering a single mode fiber: There is still intramodal dispersion (dispersion that occurs within a mode). Material dispersion arises from the variation of the refractive index of the core material as a function of wavelength.

In optical fibers above the 1300 nm wavelength, shorter wavelength light (represented by “blue” pulses) travel faster than longer wavelength light (represented by “red” pulses). Note that this situation is opposite to that of visible light, where, in physics, we learnt that “red” light travel faster than “blue” light.
This type of dispersion, where “blue” light is faster than “red” light is known as anomalous dispersion.

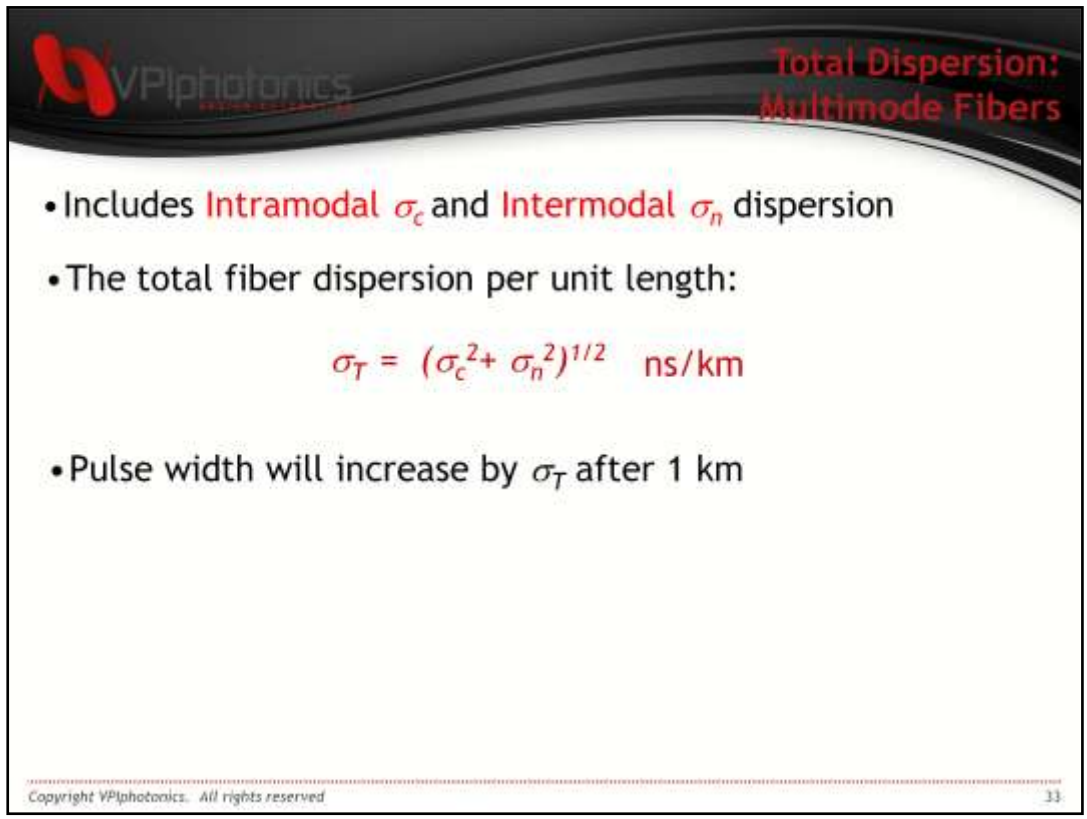
Thus, a pulse in a dispersive medium will spread out, due to the “red” components of its spectrum lagging behind the “blue” components. Similar to intermodal dispersion, intramodal dispersion arises as a result of the different speeds of the different components of a pulse of light (in this case, the different frequency components in a single mode of a pulse).



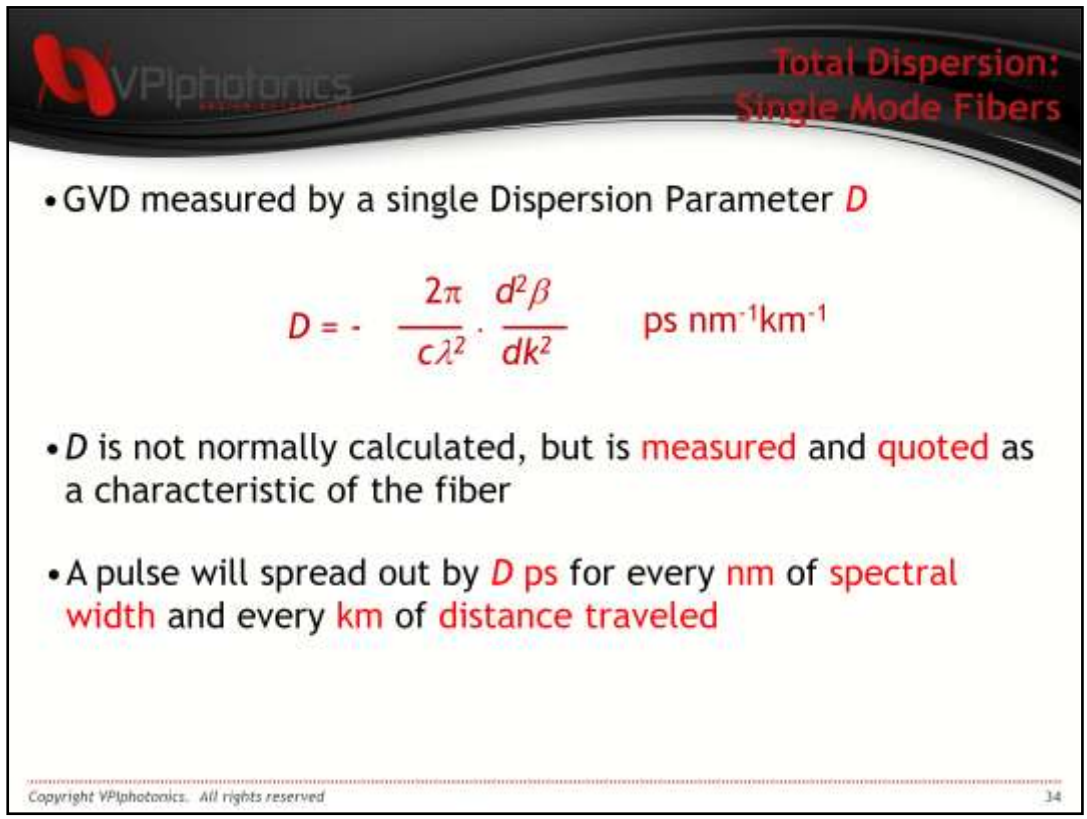
Material dispersion arises from the variation of the refractive index of the core material as a function of wavelength.



Waveguide dispersion : occurs because a single mode fiber confines only about 80 % of the optical power to the core
The velocity depends on the proportion of power in the core
At longer wavelengths the wave is less tightly confined to the core: therefore, on average, it sees a lower refractive index.



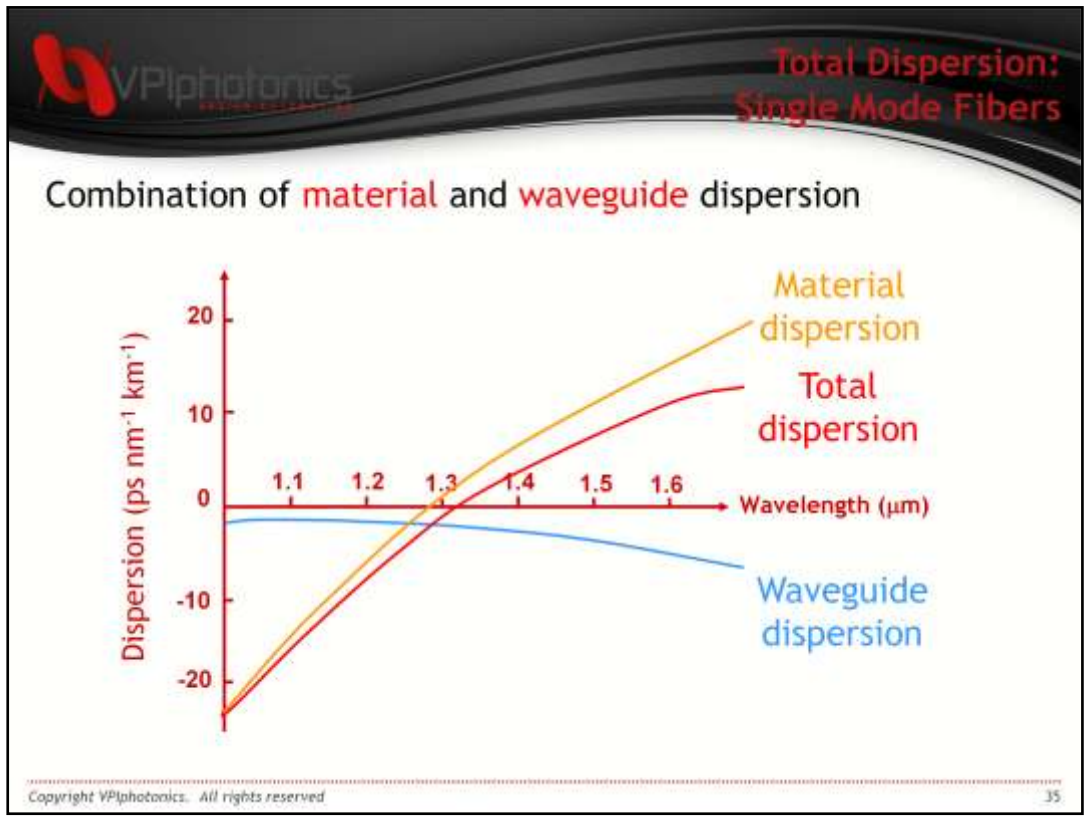
In general, for a multimode fiber – total dispersion = sum of all the different components of dispersion. Note that in multimode fibers, intermodal dispersion tends to dominate and swamp the effects of the other dispersion. Once the dominant component of dispersion is removed, the next most dominant component takes over (e.g. intramodal dispersion in single mode fibers).



The pulse broadening in single mode fibers results almost entirely from intramodal dispersion (no intermodal dispersion)
The total dispersion of single mode fiber is evaluated by GVD, which is represented by the parameter *D* in units of ps/nm/km in datasheets. Note that *D* is measured, whereas GVD (represented by an equivalent parameter $beta_2$) can be calculated.

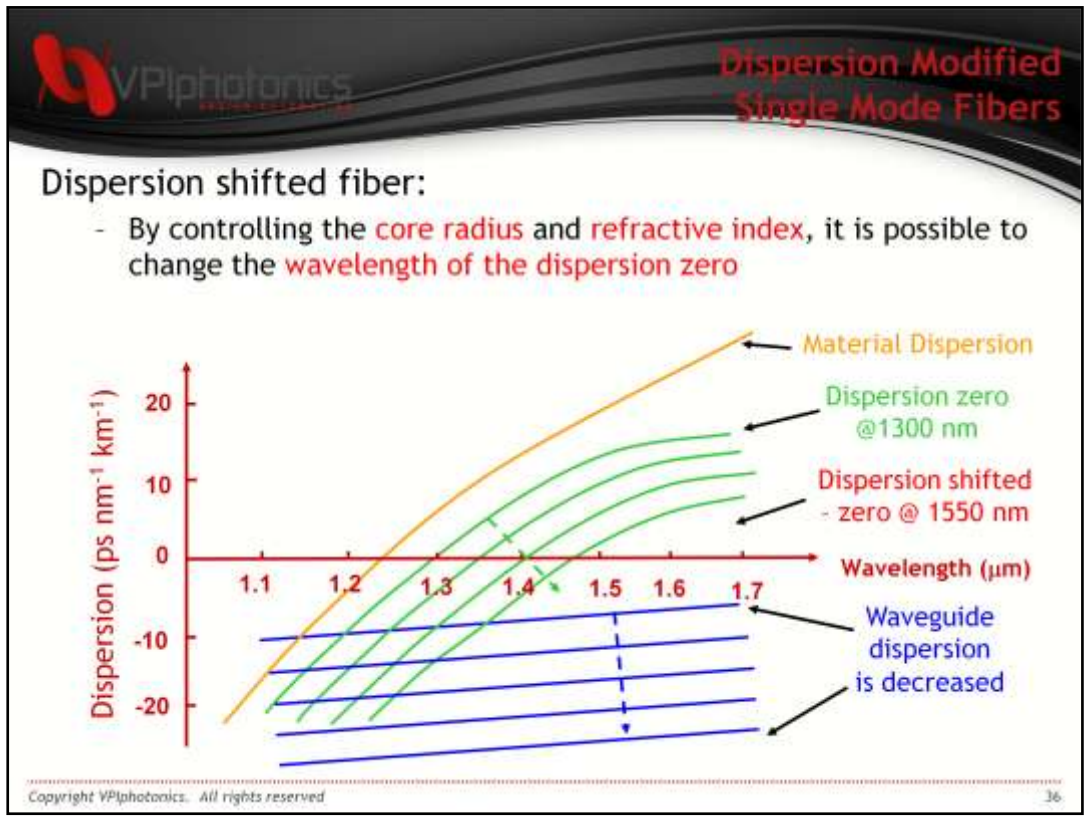
D

ps nm spectral width km distance travelled



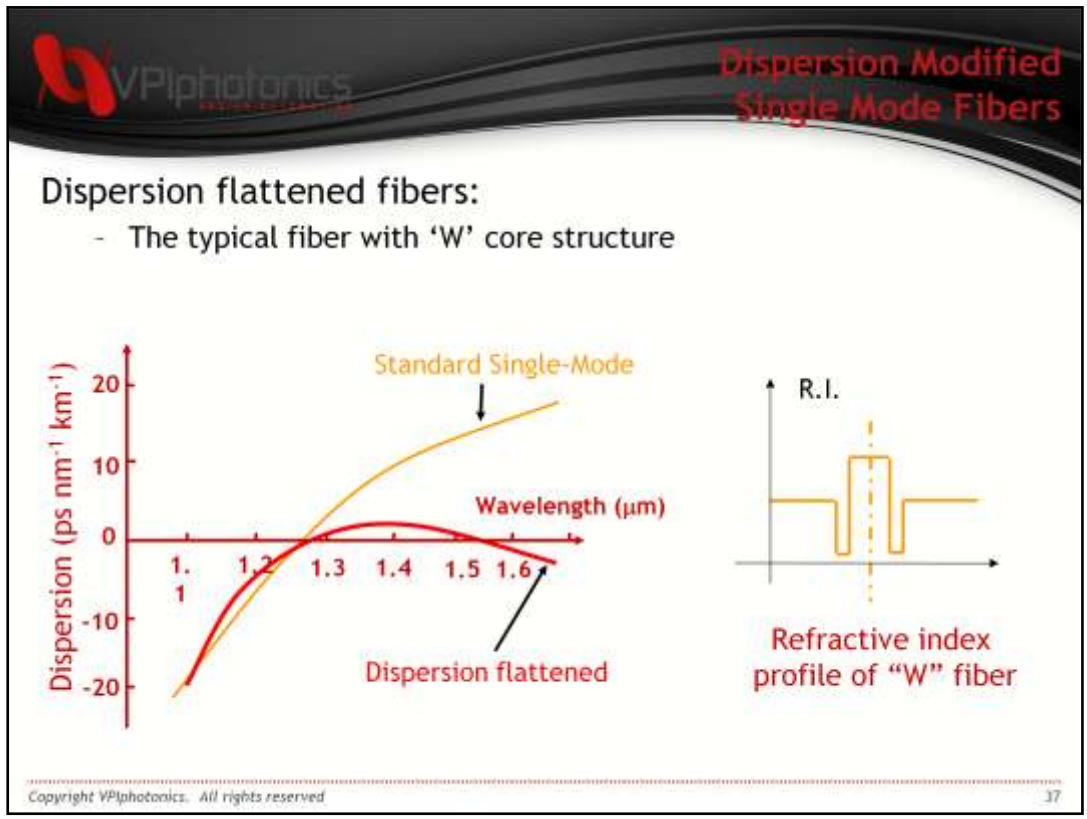
The curves of dispersion vs wavelength for the various components of intramodal dispersion is shown above. The interesting feature of the curves in the graph above is the slopes of the material (positive) and the waveguide dispersion (negative) curves. The two curves add to give the total dispersion curve.

So, what is the significance of this (slopes of the different curves)?



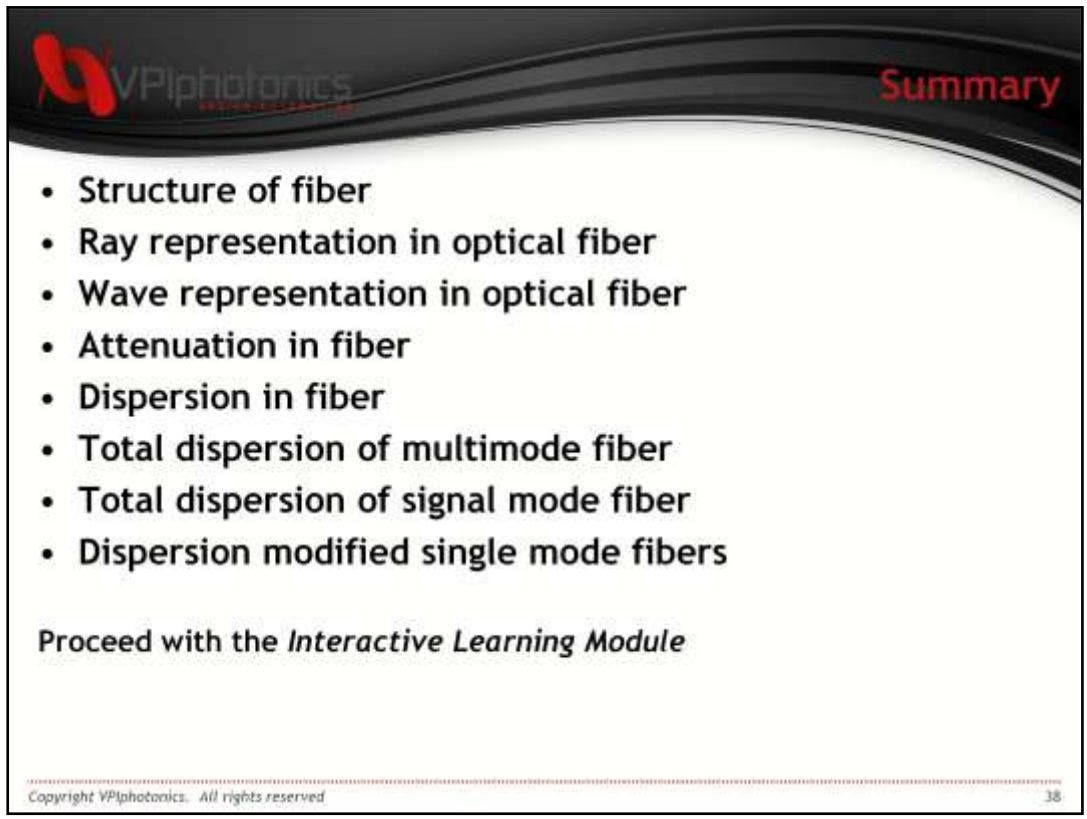
This means that the (total) dispersion curve of the fiber can be modified by changing the waveguide dispersion (i.e. changing the refractive index profile the fiber, i.e. changing the structure of the fiber). I.e. controlling the core radius, and the refractive index of the core and cladding.

This is how dispersion shifted fibers are made – the zero dispersion point of standard fibers is normally at the wavelength of 1.3 um. But for some applications, it is advantageous to shift this zero dispersion point to 1.55 um (which is also the wavelength at which fiber attenuation is at a minimum).



Even more radical modifications of the dispersion can be achieved. An example of this is dispersion flattened fibers. This is done by deliberately introducing a ring of material around the core which has a lower dispersion (known as a "depressed cladding") than the regular cladding (as shown in the refractive index profile in the slide). This is known as a "W" profile fiber, and the effect of the ring is to alter the shape the of dispersion curve, flattening the dispersion over a range of wavelength.

This has the result of reducing the dispersion slope over a wide wavelength range, and/or introduce a total dispersion curve with a negative slope (at wavelengths longer than the wavelength of the maxima of the dispersion curve). Such fibers find applications in dispersion compensation and dispersion slope compensation for example.



Summary of the key ideas of the lecture part of this module